
Adaptive Economic Aggregate Disambiguation

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Abstract

When evaluators may disagree, a decision system must choose not only how to aggregate the current panel, but whether to stop, buy another evaluation, or generate a replacement. We introduce adaptive economic aggregate disambiguation systems, a class of sequential mechanisms that make these choices under evaluation cost, delay, and decision loss. For any declared finite-horizon workload and cost model, a Bellman recursion characterizes the planner-optimal action, so no panel size or appeal rule is optimal across all tasks. We also give statistical guarantees that remain valid under adaptive stopping and changing candidate resolutions, together with finite-population bounds for uniform and weighted sampling without replacement. Synthetic examples show when further evaluation reduces total loss and when conservative bounds remain uninformative.

Keywords: aggregate disambiguation; optimal stopping; adaptive panels; finite-population sampling; optimistic democracy

1 Introduction

Finite-panel systems face two coupled decisions. They must aggregate the panel that has already voted, and they must decide whether another experiment is worth buying. The first question is statistical; the second depends on the cost of evaluation, delay, and a wrong final decision. A single panel size cannot be economically optimal for every task because both statistical difficulty and decision loss vary across transactions.

Part 1 studies when a finite panel reproduces the decision of a declared evaluator population. This paper adds adaptive resource allocation. A system may draw more evaluator episodes for the current resolution, generate a replacement resolution, or stop. We call this a *generative adaptive economic Aggregate Disambiguation System* (G-AE-ADS). Optimistic Democracy (OD) is the motivating application: it begins with a small committee and purchases larger experiments only when an external actor appeals.

The analysis separates three layers that are often conflated. A social planner chooses the action minimizing declared work, latency, and decision loss. A statistical certificate controls the risk of whatever action is selected. A decentralized mechanism must then make privately rational appeals coincide with the planner’s continuation region. The first two layers are developed here; Part 4 studies the third.

The contributions are:

1. a row–column model in which re-evaluation adds evaluator evidence while regeneration changes the candidate resolution;
2. a finite-horizon Bellman characterization and an exactly solved four-round instance;
3. selection-valid certificate composition under changing resolutions, together with uniform and weighted-without-replacement single-round bounds;
4. a deterministic work guarantee showing that the non-repeated nominal OD ladder is less than six-competitive against one oracle-known adequate panel; and
5. exact and Monte Carlo diagnostics that exercise the planner and heterogeneous finite-population sampling assumptions.

These statements do not identify the population decision with semantic truth and do not prove that the deployed appeal equilibrium is socially optimal. They instead specify the benchmark, statistical premises, and measurable quantities that such a claim would require.

Section 2 defines the decision problem and its guarantees. Section 3 reports the synthetic results, Section 4 maps them to OD, and Section 5 gives implications and future work. Formal statements and proofs are in the appendices; simulation details and secondary figures are in the supplementary material.

2 Methodology

2.1 Adaptive economic disambiguation

Fix a base problem and let P_r denote the current generated resolution at round r . A history H_r records the current resolution, observed panel outcomes, and every variable that affects a future draw. At each history the system may STOP, REEVALUATE the current resolution with another evaluator panel, or REGENERATE a new resolution. Each experiment has a transition kernel and a real incremental cost. The fixed-column model, in which only re-evaluation is available, is a strict specialization. Appendix A gives the measurable definition.

For a policy π , let P_τ and d_τ be the resolution and binary decision at its stopping time. Given a declared workload law, evaluator and generator kernels, decision loss $L(P)$, round costs c_r , and latency price λ , the social objective is

$$\mathcal{J}(\pi) = \mathbb{E} \left[\sum_{r=0}^{\tau} c_r + \lambda \tau + L(P_\tau) \mathbf{1}\{d_\tau \neq D_\infty(P_\tau)\} \right]. \quad (1)$$

Here $D_\infty(P)$ is the decision of the declared evaluator population for the resolution actually finalized. It is not a semantic truth label. If generation quality matters independently, the objective must include a fidelity loss or restrict admissible generator transitions; otherwise a planner may rationally “shop” for an easier but less faithful resolution.

2.2 Planner-optimal stopping

Let $s_r(h)$ be the conditional loss from stopping at history h , and let a_{r+1} include the work and delay caused by an available experiment. For a finite horizon, the classical Bellman recursion specialized to G-AE-ADS is

$$\begin{aligned} V_J(h) &= s_J(h), \\ V_r(h) &= \min \left\{ s_r(h), \min_{u \neq \text{STOP}} \mathbb{E}[a_{r+1}(h, u, H_{r+1}) + V_{r+1}(H_{r+1}) \mid H_r = h, u] \right\}. \end{aligned} \quad (2)$$

The system continues precisely when the best experiment has lower conditional value than stopping. The formal result and its measurability conditions are Theorem A.5. If an experiment costs $a > 0$ and reduces error probability by $\Delta e > 0$, its net value changes sign when the transaction-specific error loss crosses $a/\Delta e$. Thus neither the optimal panel nor the optimal appeal rule is workload-independent.

2.3 Selection-valid risk control

Adaptive stopping invalidates a fixed-time confidence statement unless selection is accounted for. Suppose each reached round has a certificate C_r whose conditional probability of certifying a wrong decision is at most α_r , given the history that caused the round to be reached. If an uncertified cap returns **Undetermined**, then

$$\mathbb{P}(\text{wrong binary finalization}) \leq \sum_{r=0}^J \alpha_r. \quad (3)$$

No cross-round independence and no fixed-resolution assumption are needed; see Theorem A.10. If the cap forces a binary decision, its residual uncertified error must be added.

Part 1 may also certify that all but a fraction β of a declared generator law have pointwise panel error at most δ . After history-dependent regeneration, this transfers only when the new generator law has a bounded density ratio κ_h relative to the declared law, yielding

$$\mathbb{P}(\text{panel error} \mid h) \leq \delta + (1 - \delta) \min\{1, \kappa_h \beta\}. \quad (4)$$

Without this condition, a proposer can concentrate on hard columns and marginal coverage has no conditional meaning (Proposition A.12).

2.4 One-round panel guarantees

For a finite active set, let each validator have a fixed probability p_i of agreeing with the declared population decision. Conditional on a committee, ballots are assumed independent; committee selection is a separate source of randomness. Under a uniform size- n draw without replacement, active-set mean agreement $1/2 + \gamma$, and strict majority, the two-stage Serfling–Hoeffding argument gives

$$\mathbb{P}(d \neq D_\infty) \leq \inf_{0 < t < \gamma} \left\{ e^{-2n(\gamma-t)^2} + e^{-2nt^2/(1-(n-1)/N)} \right\} \leq 2e^{-n\gamma^2/2}. \quad (5)$$

Theorem C.4 states the complete finite-population result. Theorem C.6 gives a separate valid bound for successive probability-proportional-to-weight sampling without replacement. Its relevant center is the panel-size-dependent inclusion-probability mean, not the naive one-draw stake mean. The weighted bound can be numerically weak under weight imbalance and does not automatically transfer to committees assembled through reuse, leader deletion, or mid-round replacement.

2.5 Schedule competitiveness

Let $n_0 < \dots < n_J$ be a panel ladder, $S_j = \sum_{i \leq j} n_i$, and let k be an oracle-known adequate single-panel size. Stopping at the first rung at least k costs at most

$$S_{j^*(k)} \leq \chi(N) \max\{n_0, k\}, \quad \chi(N) = \max \left\{ 1, \max_{j \geq 1} \frac{S_j}{n_{j-1}} \right\}. \quad (6)$$

For the non-repeated nominal OD sizes through 1535, $\chi = 4595/769 \approx 5.9753$ and the tight integer-grid ratio is $4595/770 \approx 5.9675$ (Theorem A.13 and Corollary A.14). This is a schedule accounting theorem: it does not reveal k , validate a stopping rule, or establish an equilibrium.

3 Results

3.1 An exactly solved adaptive planner

We solve a four-round fixed-column ladder with panel sizes (5, 7, 11, 13). The latent population decision has a symmetric prior, while persistent evaluator agreement is 0.60, 0.75, or 0.90 with probabilities 0.40, 0.40, and 0.20. A wrong final decision costs 100 units, each evaluator seat costs one unit, and latency has zero additional price. Exact backward induction uses the complete vote history for stopping, while the protocol output remains the strict majority of the most recent round.

The optimal policy purchases round 1 only after the two closest initial tallies. It dominates both immediate stopping and mechanically buying all four panels in the declared objective:

Policy	Expected seats	Error probability	Total objective
Stop after five	5.0000	0.17009	22.0094
Bellman policy	8.7067	0.09496	18.2029
Buy all rounds	36.0000	0.10127	46.1273

The full state calculation is Proposition A.8. Figure 1 recomputes the policy as decision loss varies; it is a calibrated example, not an estimate of deployed transaction values.

3.2 The cost of a resetting ladder

Figure 2 evaluates the deterministic bound in Equation (6). The worst transition is from the size-769 appeal panel to the size-1535 normal panel. Because each round discards earlier ballots, cumulative work can approach six times one oracle-known adequate panel. If the same nominal rungs accumulated evidence and bought only incremental seats, the analogous geometric envelope would be below two. The gap is therefore primarily the price of resetting ballots, not adaptivity itself.

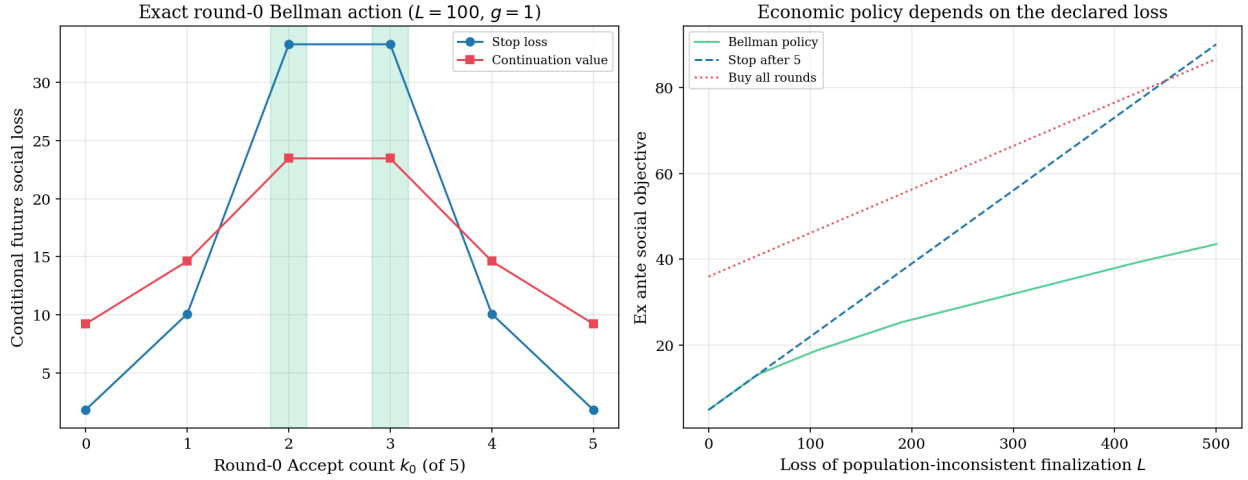


Figure 1. Exact four-round planner. *Left:* round-zero stopping and continuation values when a wrong decision costs 100; only the two closest tallies continue. *Right:* ex ante objective under a policy recomputed at every decision-loss value.

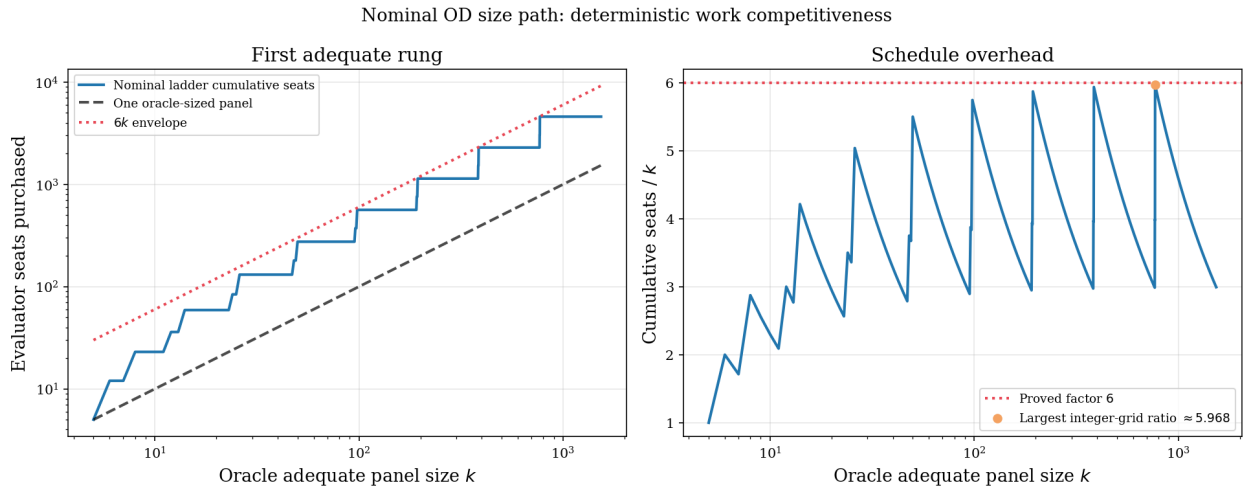


Figure 2. Evaluator-work competitiveness of the strictly increasing nominal size path through 1535. The ratio jumps when the adequate size first exceeds an appeal rung and the next normal panel is nearly twice as large. The oracle supplies the stopping point; the running protocol does not observe it.

3.3 Heterogeneous finite-population validation

The synthetic finite-population experiment fixes 199 validators: 100 have agreement probability 0.51 and 99 have probability approximately 0.891919, giving exact arithmetic mean 0.70. For each odd panel size, we compute the committee-composition law exactly, mix the corresponding Poisson–Binomial ballot tails, and compare them with Monte Carlo. Figure 3 separately checks the Serfling committee-selection stage and the complete two-stage bound.

The simulation confirms coverage, but it also exposes conservatism. Small panels can have substantial exact error while the convenient exponential cap remains equal to or above one. A valid inequality is therefore not automatically an operational certificate at the first ladder rungs.

Secondary homogeneous-policy, appeal-pressure, expected-cost, and clarity-floor diagnostics are reported in Supplement S1. They are model checks rather than estimates of live-network performance.

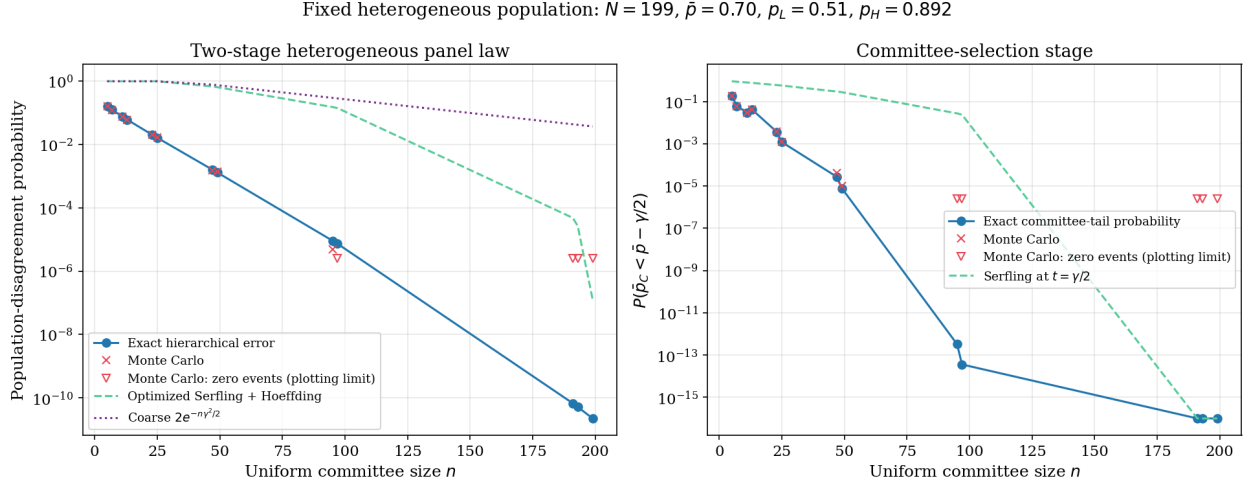


Figure 3. Exact and Monte Carlo validation under a heterogeneous finite population. *Left:* total panel error and the optimized and coarse two-stage bounds. *Right:* the probability that committee selection lowers mean agreement by more than half the active-set margin, together with Serfling’s bound. Downward triangles mark zero events in 200,000 draws.

4 Applied Case: Optimistic Democracy

OD instantiates the G-AE-ADS action geometry but not every simplifying law used for closed-form calculations. A normal leader produces a resolution, an appeal jury re-evaluates that frozen resolution, and a successful challenge can return the transaction to proposing under a new leader. Re-evaluation therefore moves down an evaluator column; regeneration moves horizontally to a new generator column. The population-reference decision must be attached to the resolution currently under consideration rather than treated as fixed across all transitions.

The nominal size sequence analyzed here is

$$5, 7, 11, 13, 23, 25, \dots, 767, 769, 1535.$$

Every panel uses fresh ballots and strict majority in the binary Accept-versus-non-Accept projection. The exact schedule and protocol abstraction are given in Appendices D–E. The less-than-six result applies to this strictly increasing, non-repeated path stopped by an oracle at its first adequate rung. It does not cover repeated capacity rounds, voided appeals, strategic over-appeal, or a history-dependent realized size path.

4.1 What transfers to the implementation

The pinned validator selector repeatedly samples proportionally to stake while rejecting duplicate or ineligible identities. Conditional on a fixed eligible set and weights, this matches the one-round successive weighted design in Theorem C.6. The theorem therefore supplies a legitimate analytical route, but not a complete deployment guarantee: its inclusion-probability margin is unobserved, its worst-case bound is weak at much of the nominal ladder, and successor committees may reuse validators, delete the previous leader, or replace idle seats.

Other implementation features also matter:

- a normal-round leader is a distinguished voting seat rather than an exchangeable evaluator;
- successful transitions can change the generated resolution;
- available validators and consecutive failed appeals make realized sizes path-dependent;
- commit-phase replacement makes the realized committee differ from the initially sampled one;
- appeal windows, explicit finalization calls, and timeouts introduce clock time; and
- multiple potential appellants create an endogenous strategic process.

These are transfer conditions, not hidden assumptions. Appendix D.6 records the full abstraction boundary.

4.2 Small-panel power and leader asymmetry

At panel size five, a homogeneous agreement probability of 0.70 gives exact majority error 16.3%; the coarse exponential bound is vacuous. If one of the five seats always agrees with its own proposal, the error becomes asymmetric: it falls to 8.4% when the population target accepts but rises to 34.8% when the target rejects with ordinary-seat acceptance probability 0.30 (Proposition E.6). The contract includes the leader in the voting array, but the deterministic self-agreement model is a sensitivity benchmark rather than a measured ballot law.

The small real-LLM pilot from Part 1 certified none of its frozen columns at panel sizes five or seven under its predeclared stringent settings; its first configured mass-bound crossing occurred at 47. This is a statement about finite-sample power on that diagnostic catalogue, not an estimate of deployed error and not evidence that uncertified tasks are wrong.

4.3 Planner action versus private appeal

For a private appellant who loses cost c after an unsuccessful challenge and earns net reward b after success, bond-only break-even requires a believed success probability above $c/(b + c)$. When the successful return is 1.5 times the posted amount, $b = c/2$ and the private threshold is $2/3$; Appendix K gives the formal reduced model. The social continuation rule is instead Equation (2), which depends on transaction loss, evaluator work, latency, and the value of information. Equality of these two regions is an implementation criterion, not a consequence of prefunding the next panel.

5 Discussion

The central implication is that economic adequacy is necessarily conditional. Statistical clarity determines how much evidence is useful, but the decision to buy it also depends on the real cost of evaluation and delay and on the loss caused by a population-inconsistent result. A high-value transaction and a self-created low-impact transaction can therefore rationally have different continuation regions even when their observed tallies are identical.

The schedule result gives a useful but narrow efficiency statement. If an oracle reveals the first adequate rung, the nominal OD ladder pays less than six times one adequate panel. It does not show that an appellant observes that rung or that the deployed process stops there. The comparison also makes a design cost visible: resetting ballots changes a roughly two-competitive geometric enlargement into a nearly six-competitive cumulative path.

The statistical results and the planner solve different problems. Selection-valid certificates control finalization risk after adaptive histories, while Bellman optimization determines whether the expected reduction in loss justifies another experiment. A mechanism can be statistically valid but economically wasteful, or economically attractive under a misspecified fidelity objective. Conversely, prefunding a panel does not guarantee that someone appeals precisely when its social value is positive.

The most important next steps are empirical and mechanism-theoretic. A deployment study should simulate the actual weighted selector over validator snapshots, estimate joint stake–performance structure, and include leader behavior, reuse, replacement, and shared execution state. The planner should be calibrated to declared transaction-loss classes and wall-clock costs rather than a universal damage number. The decentralized layer then needs a preemption-aware multi-appellant equilibrium, capital constraints, and a reward rule whose private continuation region approximates the calibrated social one. Generator fidelity must also be priced or constrained before regeneration can be interpreted as welfare-improving.

Part 1 provides finite-panel and generator-coverage certificates. This paper turns those statistical objects into adaptive experiments and a planner benchmark. Part 4 asks whether private appeals implement that benchmark. Together they support a precise conditional claim: OD can be evaluated as an adaptive economic disambiguation mechanism once the workload loss, experiment kernels, sampling design, and appellant incentives are declared. They do not yet establish optimality of the deployed network.

Institutional Disclosure

This work was conducted within GenLayerLabs Research. GenLayerLabs develops GenLayer, whose nominal Optimistic Democracy panel ladder is analyzed as an application case. The general statistical and decision-theoretic results do not depend on that implementation.

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Appendices: Proofs and Technical Results

A Adaptive Economic Aggregate Disambiguation

A.1 Definition

Fix a base problem X . Part 1 associates every generator configuration $a \in \mathcal{A}$ with a frozen resolution $T_a = \text{Solve}(a, X)$ and therefore with a column instance $P(a) = (X, T_a)$ and population-reference decision $D_\infty(P(a))$. An OD execution can spend work in two directions: draw more evaluator rows against the current column, or replace that column through a new generation event.

Definition A.1 (Generative adaptive economic ADS). A *generative adaptive economic Aggregate Disambiguation System* (G-AE-ADS) consists of:

1. a declared class $\mathcal{P}_X$ of Part 1 column instances and an initial generator kernel;
2. a finite horizon J , observable histories H_r , and a current column $P_r \in \mathcal{P}_X$ recorded in H_r ;
3. at each nonterminal history h , an action set $\mathcal{U}_r(h)$ containing STOP and any available experiments, such as REEVALUATE or REGENERATE;
4. for every experiment u , a transition kernel $Q_{r+1}^u(dP', do \mid h)$ jointly producing the next column P' and its evaluator observation o ; REEVALUATE is supported on $P' = P_r$, whereas REGENERATE may draw a new column;
5. a provisional binary decision $d_r = \psi_r(H_r)$, a nonnegative real resource cost for each experiment, and a terminal loss.

A policy is history-measurable, chooses one action after each observation, and must stop by J . The output at stopping time τ is (P_τ, d_τ) . A declared Undetermined action may be included with its own loss.

The history may contain generator identities, frozen resolutions, eligible validators, past committees, private watcher signals, and protocol status. These variables cannot be removed when they affect a future kernel. A panel size ladder is encoded by restricting the experiments available at each history and assigning their evaluator-seat costs.

Definition A.2 (Fixed-column AE-ADS). The *fixed-column AE-ADS* is the specialization in which $P_r = P_0$ almost surely for every round and the only nonterminal experiment is REEVALUATE. It has one target $D_\infty(P_0)$ throughout the execution.

Each evaluator round remains one finite-panel Part 1 ADS experiment. The deployed appeal flow is generative: an appeal jury re-evaluates a frozen proposal, while a successful appeal or another return to proposing can create a new column. The fixed-column model used for several closed-form bounds below is therefore a transparent specialization, not a description of every OD transition. Neither specialization makes a population decision semantic truth.

A.2 Risk, cost, and the meaning of economic optimality

Let Λ be a declared law over base problems, generator transitions, evaluator kernels, and any latent variables needed by the execution. For a policy π , define the final population-consistency risk and expected real resource cost by

$$R_\Lambda(\pi) = \mathbb{P}_\Lambda(d_\tau \neq D_\infty(P_\tau)), \quad C_\Lambda(\pi) = \mathbb{E}_\Lambda \left[\sum_{r=0}^{\tau} c_r \right].$$

Definition A.3 (Economic objective and constrained optimum). Let $L(P) \geq 0$ be the social loss of a population-inconsistent binary decision and let $\lambda \geq 0$ price one round of latency. The Bayes social objective is

$$\mathcal{J}_\Lambda(\pi) = \mathbb{E}_\Lambda \left[\sum_{r=0}^{\tau} c_r + \lambda \tau + L(P_\tau) \mathbf{1}\{d_\tau \neq D_\infty(P_\tau)\} \right].$$

Alternatively, for a model class $\mathfrak{L}$ and target δ , define the hard-risk feasible set

$$\Pi_\delta = \left\{ \pi : \sup_{\Lambda \in \mathfrak{L}} R_\Lambda(\pi) \leq \delta \right\},$$

and minimize expected resource cost over Π_δ . An “economically optimal” policy is always relative to one of these declared objectives and an admissible policy class.

Remark A.4 (Population consistency does not price generation quality). The consistency target in Definition A.3 belongs to the resolution actually finalized; it is a declared population decision, not semantic truth. Consequently, a REGENERATE policy can reduce the displayed inconsistency term by moving to a clearer column even when that resolution is less faithful to the original problem. When the reduction in expected inconsistency loss is large relative to the regeneration cost, this “clarity shopping” is rational for the declared planner. It is an objective- specification issue, not a failure of the panel certificate or of Proposition A.12.

A welfare interpretation that values generation quality must therefore either add a terminal fidelity loss comparing P_τ with the original problem, or restrict the admissible generator kernels to a predeclared semantic-fidelity class. The fixed-column model is the strongest such restriction. Part 4’s implementation regret is relative to whichever benchmark is declared here and cannot repair a missing fidelity term. Finally, a bond returned by one participant to another is a transfer in the social objective; generation, evaluation, latency, burns, and external decision losses are real costs. Private appeal incentives need not minimize $\mathcal{J}_\Lambda$.

A.3 The planner’s stopping rule

Under a declared prior and likelihood, history h induces a posterior over the current column, its population decision, and future generator/evaluator transitions. Write

$$s_r(h) = \mathbb{E}[L(P_r)\mathbf{1}\{d_r \neq D_\infty(P_r)\} \mid H_r = h]$$

for the conditional loss from stopping at round r . Fold latency and all real incremental resources into an integrable continuation cost $a_{r+1}(h, u, H_{r+1}) \geq 0$ for experiment u . Assume throughout this subsection that the column, observation, action, and history spaces are standard Borel; that $P \mapsto D_\infty(P)$, all transition kernels, decision maps, action correspondences, and costs are measurable; and that terminal and experiment costs are integrable under the declared law. These conditions provide regular conditional distributions and measurable Bellman values. For simplicity, each reachable action set below is finite; standard measurable-selection conditions give the corresponding result for general action spaces. Additional global constraints must be included in the state. The following is the classical finite-horizon dynamic-programming principle [2], formulated here for the AE-ADS state and social objective. Its role is to identify the planner benchmark that a distributed appeal mechanism would have to implement.

Theorem A.5 (Finite-horizon economically optimal action). *For the Bayes objective in Definition A.3 and a G-AE-ADS with finite horizon J , the minimum conditional future social loss, excluding costs already sunk at the current history, satisfies*

$$\begin{aligned} V_J(h) &= s_J(h), \\ V_r(h) &= \min \left\{ s_r(h), \min_{u \in \mathcal{U}_r(h) \setminus \{\text{STOP}\}} \mathbb{E}[a_{r+1}(h, u, H_{r+1}) + V_{r+1}(H_{r+1}) \mid H_r = h, u] \right\} \end{aligned}$$

for $r < J$. An optimal policy chooses an experiment attaining the inner minimum when its value is strictly smaller than $s_r(h)$ and stops when the reverse inequality holds, with arbitrary randomization among ties.

Proof. At the terminal round, stopping is mandatory and incurs s_J . Suppose the formula gives the optimal value at round $r + 1$. At history h in round r , every admissible policy must either stop, incurring $s_r(h)$, or choose one available experiment, pay its cost, and thereafter incur at least $V_{r+1}(H_{r+1})$. Minimizing over these alternatives is optimal. Backward induction proves the recursion and the stated policy. $\square$

In the fixed-column two-action specialization, re-evaluation is socially rational when its expected value of information exceeds its incremental cost:

$$s_r(h) - \mathbb{E}[V_{r+1}(H_{r+1}) \mid h, \text{REEVALUATE}] > \mathbb{E}[a_{r+1}(h, \text{REEVALUATE}, H_{r+1}) \mid h, \text{REEVALUATE}].$$

Corollary A.6 (No workload-free economic optimum). *Suppose that at some reachable history an experiment has positive expected cost $a > 0$ and reduces conditional terminal error probability by $\Delta e > 0$, holding other losses fixed. If a population-inconsistent decision has loss L , stopping is preferred when $L < a/\Delta e$, whereas the experiment is preferred when $L > a/\Delta e$. Consequently no appeal policy is economically optimal independently of the workload law, error loss, resource prices, and latency price.*

Proof. The net reduction in conditional social loss from continuation is $L\Delta e - a$. Its sign changes at $L = a/\Delta e$. $\square$

Remark A.7 (Distributed implementation and Part 4). Theorem A.5 is a planner benchmark, not a description of the deployed appeal equilibrium. OD implements the same action only if, at every reachable history, at least one actor finds it privately profitable to trigger the socially preferred available experiment—and refrains when stopping is preferred. Bonds can fund the next round yet still underprovide appeals through free-riding or overprovide them when private transaction value exceeds social value. Part 4 studies this implementation gap.

A.4 A solved planner instance

The Bellman equation becomes operational once its primitives are declared. Consider the fixed-column four-round ladder $\mathcal{N} = (5, 7, 11, 13)$. Let the latent population decision $D \in \{0, 1\}$ have the symmetric prior, and let a task's persistent reference-agreement probability take values

$$p \in \{0.60, 0.75, 0.90\} \quad \text{with probabilities} \quad (0.40, 0.40, 0.20),$$

independently of D . Conditional on (D, p) , rounds are independent and an Accept ballot has probability $q_{D,p} = p$ when $D = 1$ and $1 - p$ when $D = 0$. The protocol output at a stop is still the strict majority of the *last* round; the planner may use the entire public history to decide whether to buy the next round.

Let $A_r = \sum_{j=0}^r k_j$ and $N_r = \sum_{j=0}^r n_j$. Binomial coefficients cancel from the likelihood ratio, so the posterior is

$$\Pi_r(d, p \mid A_r) = \frac{\rho(d, p) q_{d,p}^{A_r} (1 - q_{d,p})^{N_r - A_r}}{\sum_{d', p'} \rho(d', p') q_{d',p'}^{A_r} (1 - q_{d',p'})^{N_r - A_r}}.$$

Consequently (r, A_r, d_r) is a sufficient Bellman state. With loss L for a population-inconsistent final decision, evaluator-seat price g , and zero latency price, the exact recursion is

$$s_r(A_r, d_r) = L \sum_p \Pi_r(1 - d_r, p \mid A_r),$$

$$V_r(A_r, d_r) = \min \left\{ s_r(A_r, d_r), g n_{r+1} + \sum_{k=0}^{n_{r+1}} m_{r+1}(k \mid A_r) V_{r+1}(A_r + k, d(k)) \right\},$$

where m_{r+1} is the posterior predictive Binomial mixture and $d(k) = \mathbf{1}[k > n_{r+1}/2]$; at $r = 3$, $V_3 = s_3$.

Proposition A.8 (Exact calibrated policy). *For the preceding instance with $L = 100$ and $g = 1$, the optimal round-zero action is to buy round 1 exactly after tallies $k_0 \in \{2, 3\}$. Its ex ante expected evaluator-seat count is 8.7067, its population-disagreement probability is 0.09496, and its objective is 18.2029. Immediate stopping has $(\mathbb{E}[G], R, \mathcal{J}) = (5, 0.17009, 22.0094)$, while mechanically buying all four rounds has $(36, 0.10127, 46.1273)$.*

Proof. Evaluate the finite recursion above by backward induction over the reachable integer states. At round zero the stop and continuation values are

k_0	0	1	2	3	4	5
stop	1.8055	10.0695	33.2975	33.2975	10.0695	1.8055
continue	9.2098	14.6123	23.4671	23.4671	14.6123	9.2098

Thus exactly the tallies $k_0 = 2$ and $k_0 = 3$ continue. Integrating the recursively selected actions against the prior predictive count laws gives the reported ex ante quantities. The two benchmarks follow from the mixture of the exact $n = 5$ and $n = 13$ Binomial tails, respectively. $\square$

Proposition A.9 (OD actions as matrix moves). *Let the Part 1 matrix have generator columns T_a and evaluator rows. In the G-AE-ADS abstraction, an appeal jury draws additional evaluator episodes for the current T_a and hence moves down one column; a return to proposing draws a new generator configuration and hence moves to another column. Therefore the applicable population reference at round r is $D_\infty(P_r)$, while the fixed-column ladder is the special path with no horizontal moves.*

Proof. The REEVALUATE kernel in Definition A.1 is supported on $P_{r+1} = P_r$, so only the evaluator observation changes. A REGENERATE kernel may instead place positive mass on a different column. The target attached to the stopped output is therefore the Part 1 population decision of its current column, $D_\infty(P_r)$. $\square$

This identification is structural rather than statistical. It does not make the realized generator transitions i.i.d., nor does it make evaluator episodes independent. Proposition A.12 states the extra condition needed to transfer Part 1's generator coverage after a horizontal move.

A.5 Selection-valid stopping certificates

Economic stopping may deliberately trade error against cost. A design with a hard risk budget instead needs certificates that remain valid after adaptive selection.

Theorem A.10 (Family-valid certified stopping). *For every reachable round r , let $C_r \in \{0, 1\}$ be a certificate measurable after observing H_r . Suppose that, almost surely on reaching round r ,*

$$\mathbb{P}_\Lambda(C_r = 1, d_r \neq D_\infty(P_r) \mid H_{r-1}, \tau \geq r) \leq \alpha_r$$

*for every $\Lambda \in \mathfrak{L}$ (with the conditioning omitted at $r = 0$). If the system stops with a binary decision only when $C_r = 1$, continues otherwise, and emits **Undetermined** rather than an uncertified binary decision at the cap, then*

$$\sup_{\Lambda \in \mathfrak{L}} \mathbb{P}_\Lambda(\text{wrong binary finalization}) \leq \sum_{r=0}^J \alpha_r.$$

No independence across rounds and no fixed-column assumption are required.

Proof. A wrong binary finalization occurs at one of the disjoint stopping rounds and requires both $C_r = 1$ and $d_r \neq D_\infty(P_r)$. The conditional premise bounds the probability contributed by each reachable round by α_r ; summing over rounds proves the claim. $\square$

Remark A.11 (Forced binary terminal output). If the cap forces an uncertified binary output, add the residual term

$$\mathbb{P}_\Lambda(\tau = J, C_J = 0, d_J \neq D_\infty(P_J)).$$

Part 1's operational certificate uses a separate predeclared generator–evaluator matrix for future independent panels. A live OD tally does not become a C_r satisfying this theorem merely by reusing that name; selection-valid calibration must be proved for the actual history and sampler.

The distinction between a fresh generator column and a challenged column is quantifiable. For a panel size K , write

$$\mathcal{G}_{K,\delta} = \{a : e_{X,K}(T_a) \leq \delta\}.$$

Part 1's population-coverage statement is $\rho_G(\mathcal{G}_{K,\delta}) \geq 1 - \beta$ for a declared generator law ρ_G .

Proposition A.12 (Generator-shift transfer). *At a reachable history h , suppose a REGENERATE action draws a new configuration from a conditional law $\rho_h \ll \rho_G$ satisfying*

$$\left\| \frac{d\rho_h}{d\rho_G} \right\|_\infty \leq \kappa_h,$$

then uses an independent K -evaluator panel with pointwise error $e_{X,K}(T_a)$. If $\rho_G(\mathcal{G}_{K,\delta}) \geq 1 - \beta$, then

$$\mathbb{P}(d \neq D_\infty(P(a)) \mid h, \text{REGENERATE}) \leq \delta + (1 - \delta) \min\{1, \kappa_h \beta\}.$$

For a fresh draw from the declared law, $\kappa_h = 1$ and this is Part 1's deployment-error bound.

Proof. Absolute continuity gives $\rho_h(\mathcal{G}_{K,\delta}^c) \leq \kappa_h \rho_G(\mathcal{G}_{K,\delta}^c) \leq \kappa_h \beta$, clipped at one. Integrate the pointwise panel error, using the bound δ on $\mathcal{G}_{K,\delta}$ and the trivial bound one on its complement. $\square$

The proposition exposes the failure mode of adaptive column selection. A proposer or transition rule can concentrate on hard columns, and conditioning on an earlier close or disputed vote can make the challenged-column law singular relative to the original generator law. In that case a marginal coverage number $R_{X;K,\delta}$ supplies no conditional bound. One must either bound κ_h , certify the realized column pointwise, or establish the history-uniform premise of Theorem A.10. If such a premise is available at every potentially finalizing row-column state, its error budgets compose even though normal successors replace columns and appeals select unusually difficult ones.

A.6 How much work does a geometric ladder waste?

The next result isolates the schedule from the stopping problem. Let $k_\delta(P)$ be the smallest single-panel size declared adequate for instance P under a chosen one-panel risk criterion whose guarantee is monotone in panel size. It is an oracle benchmark: $k_\delta(P)$ is generally unknown to the running protocol.

Theorem A.13 (Ladder competitiveness against an adequate panel). *Let $0 < n_0 < \dots < n_J$ be a panel ladder and $S_j = \sum_{i=0}^j n_i$. Define its cumulative schedule coefficient*

$$\chi(\mathcal{N}) = \max \left\{ 1, \max_{1 \leq j \leq J} \frac{S_j}{n_{j-1}} \right\}.$$

If $j^(k) = \min\{j : n_j \geq k\}$, then for every $k \in (n_0, n_J]$,*

$$S_{j^*(k)} < \chi(\mathcal{N}) k.$$

For $0 < k \leq n_0$, the incurred work is n_0 . Consequently, for every $0 < k \leq n_J$,

$$S_{j^*(k)} \leq \chi(\mathcal{N}) \max\{n_0, k\}.$$

Thus a ladder stopped at the first oracle-adequate rung is $\chi(\mathcal{N})$ -competitive in cumulative evaluator seats against the startup-aware oracle benchmark $\max\{n_0, k\}$; equivalently, it is $\chi(\mathcal{N})$ -competitive against one panel of size k whenever $k > n_0$.

Proof. For $j = j^*(k) \geq 1$, minimality gives $n_{j-1} < k \leq n_j$. Hence $S_j \leq \chi(\mathcal{N}) n_{j-1} < \chi(\mathcal{N}) k$. When $j = 0$, $S_0 = n_0 \leq \chi(\mathcal{N}) n_0$, because $\chi(\mathcal{N}) \geq 1$. This also proves the unified bound. $\square$

Corollary A.14 (Nominal OD size path is less than six-competitive). *Consider the strictly increasing, non-repeated nominal size sequence through the first capacity round,*

$$5, 7, 11, 13, 23, 25, \dots, 767, 769, 1535.$$

If an oracle stops it at the first rung at least $k \leq 1535$, its cumulative evaluator-seat work is less than

$$6 \max\{5, k\}.$$

The schedule coefficient in Theorem A.13 is $4595/769 \approx 5.9753$. Because k is integer, the tight realized worst-case ratio is slightly smaller: $4595/770 \approx 5.9675$.

Proof. Write the normal and appeal sizes as $m_j = 6 \cdot 2^j - 1$ and $a_j = 6 \cdot 2^j + 1$. Cumulative work through m_j is $18 \cdot 2^j - 13$, whose ratio to the preceding a_{j-1} is less than 6. Cumulative work through a_j is $24 \cdot 2^j - 12$, whose ratio to the preceding m_j is less than 4. The largest finite ratio occurs at the final transition from 769 to 1535 and equals $4595/769$. Apply Theorem A.13. For integer k , a transition past 769 first occurs at $k = 770$; direct substitution gives $4595/770$. The analogous normal-transition ratios increase with the rung, while the appeal-transition ratios remain below four, so this is the realized maximum. $\square$

Corollary A.15 (Workload adaptation). *If $k_\delta(P) \leq 1535$ almost surely under a workload law Λ , the oracle-stopped nominal ladder satisfies*

$$\mathbb{E}_{P \sim \Lambda}[S_{j^*(k_\delta(P))}] < 6 \mathbb{E}_{P \sim \Lambda}[\max\{5, k_\delta(P)\}].$$

A fixed panel adequate for every instance in the workload instead pays its chosen worst-case size $k_{\max} = \text{ess sup}_\Lambda k_\delta(P)$ on every instance. In particular, the displayed bound itself certifies an expected-work improvement whenever $6\mathbb{E}[\max\{5, k_\delta(P)\}] < k_{\max}$; outside that sufficient regime the actual workload comparison must be computed rather than inferred from the word “adaptive.”

Remark A.16 (What the factor six does and does not say). The factor is an accounting property of the schedule, including the fact that earlier ballots are discarded. If the same nominal rungs accumulated evidence and purchased only the incremental seats, reaching the first rung at least k would pay just that final rung, which is less than $2 \max\{5, k\}$ on this sequence. Thus most of the factor-six envelope is the price of resetting rather than accumulating ballots. It does not show that an appellant observes $k_\delta(P)$, that current OD stops at the first adequate rung, that the risk criterion is attainable under the deployed sampler, or that six is minimax over all sequential designs. Repeated capacity rounds, voided appeals, and strategic over-appeal add work outside the bound. Theorem A.5 specifies the economic stopping target; Theorem A.10 specifies one route to hard-risk validity; Part 4 asks whether decentralized incentives implement either rule.

B The Validator Population

B.1 Reference Decision Imported from Part 1

We import Part 1’s population decision, then add a hierarchical layer that is specific to this paper. It is important not to identify that new layer with Part 1’s conditional finite-census regime: Part 1 conditions on a realized binary census, whereas here an active validator retains a stochastic ballot probability.

Part 1 defines an interpreter population as a probability space $(\Theta, \mathfrak{G}, \nu)$ with a jointly measurable evaluation map; for a fixed task, each index θ has a measurable acceptance probability $q(\theta)$, the population distribution is the pushforward $F = q_*\nu$ on $[0, 1]$, and the limit decision is

$$D_\infty = \mathbf{1}[\mu \geq \tfrac{1}{2}], \quad \mu = \mathbb{E}_\nu[q].$$

D_∞ is thus defined by the declared population measure alone—*before* any committee votes. It is a population decision, not an external truth label. For this part only, define a validator’s probability of agreeing with that reference by

$$p(\theta) = \mathbb{P}(V_\theta = D_\infty) = \begin{cases} q(\theta) & \text{if } D_\infty = 1, \\ 1 - q(\theta) & \text{if } D_\infty = 0, \end{cases}$$

a measurable function of θ . We call $p(\theta)$ a *reference-agreement probability*; calling it semantic accuracy would add a ground-truth premise that Part 1 explicitly does not supply.

B.2 The Finite Active Set

The deployed system does not sample committees from the infinite population directly. At each epoch, a finite **active validator set**

$$S_N = \{\theta_1, \dots, \theta_N\}, \quad \text{agreement probabilities } p_i = p(\theta_i),$$

is fixed (validators who joined, staked, and were primed). Committees are then sampled from S_N . Throughout this part we condition on these validator identities and probabilities, so the p_i are fixed constants, and write

$$\bar{p} = \frac{1}{N} \sum_{i=1}^N p_i, \quad \sigma_p^2 = \frac{1}{N} \sum_{i=1}^N (p_i - \bar{p})^2, \quad \gamma = \bar{p} - \tfrac{1}{2}.$$

We assume $\gamma > 0$: the active set is, on average, more likely than not to reproduce the declared population decision. This γ is the *active-set agreement margin*. It equals Part 1’s population clarity only in the corresponding i.i.d. superpopulation limit; for a fixed epoch it is a separate, unobserved parameter. If the active set is drawn i.i.d. from ν , then $\bar{p} \rightarrow \mathbb{E}_\nu[p] = 1/2 + |\mu - 1/2|$ almost surely away from the threshold.

Remark B.1 (Stake weighting). The implemented selector repeatedly draws proportionally to stake and rejects duplicates or ineligible identities. Conditional on a fixed then-eligible set, this is successive weighted sampling without replacement. Its sample mean is not centered at the naive one-draw stake mean: first-order inclusion probabilities depend on panel size and are generally not n times the drawing weights [7]. Theorem C.6 below gives a one-round bound for that design. Stake used to define the declared evaluator population would instead change the target itself; that modeling choice remains distinct from the committee-selection law.

Remark B.2 (Independence within a committee). Given a committee, we *assume* conditionally independent ballot episodes with parameters (p_i) . Part 1 justifies this under a product law for private evaluator inputs; independent seeds alone do not remove dependence caused by shared random external state. Committee members are selected without replacement within a committee, appeal juries exclude consumed validators, and normal successors may reuse eligible survivors although every successor resets its ballots. Section C.1 handles one uniform or successive-weighted draw; shared-state dependence and history-dependent survivor/replacement effects remain outside a one-round abstraction unless its conditional premises hold uniformly.

C Conditional Ballots and Committee Sampling

Conditional on a committee C , the reference-agreement count $S_C = \sum_{i \in C} A_i$ is Poisson–Binomial under the independent-ballot assumption, with

$$\mathbb{E}[S_C \mid C] = \sum_{i \in C} p_i, \quad \text{Var}(S_C \mid C) = \sum_{i \in C} p_i(1 - p_i).$$

Writing $\bar{p}_C = n^{-1} \sum_{i \in C} p_i$, Hoeffding gives the only conditional fact needed below:

$$\mathbb{P}\left(\frac{S_C}{n} - \bar{p}_C \leq -t \mid C\right) \leq e^{-2nt^2}. \quad (7)$$

At fixed committee mean, heterogeneity reduces the count variance relative to a Binomial with the same mean, but this observation neither preserves the mean under unequal-probability sampling nor handles dependent ballots.

C.1 Uniform sampling without replacement

The next layer randomizes the committee itself. We begin with the uniform design, then give a separate theorem for probability-proportional-to-weight successive sampling. Appeal-jury exclusions can be represented by changing the eligible population before a draw; multi-round adaptive reuse requires history-uniform premises.

Definition C.1 (Committee draw). A committee of size n is a uniformly random size- n subset $C \subseteq S_N$ drawn without replacement. Its mean agreement probability is the random variable

$$\bar{p}_C = \frac{1}{n} \sum_{i \in C} p_i, \quad \mathbb{E}[\bar{p}_C] = \bar{p}.$$

At later model rounds the eligible set may be replaced by $S_N \setminus (\text{consumed validators})$, provided the next committee is uniform on that set. This definition is not a claim about the deployed stake-weighted sampler.

For sampling without replacement, the correct Hoeffding-type bound is Serfling’s [4]:

Theorem C.2 (Serfling). *Let $(x_1, \dots, x_N)$ be a finite population with values in $[0, 1]$ and mean $\bar{x}$, and let $X_1, \dots, X_n$ be drawn without replacement. Then for every $t > 0$:*

$$\mathbb{P}\left(\frac{1}{n} \sum_{j=1}^n X_j - \bar{x} \geq t\right) \leq \exp\left(-\frac{2nt^2}{1 - f_n^*}\right), \quad f_n^* = \frac{n-1}{N},$$

and symmetrically for the lower tail.

Since $1 - f_n^* \leq 1$, the bound is *at least as strong* as the with-replacement Hoeffding bound. If $n = N$, the sample mean equals the finite-population mean exactly, so committee-draw error is zero. This statement concerns uniform sampling from the entire modeled population, not the protocol's terminal all-eligible round.

Corollary C.3 (Committee-mean concentration). *For a committee of size n drawn as in Definition C.1:*

$$\mathbb{P}(|\bar{p}_C - \bar{p}| \geq t) \leq 2 \exp\left(-\frac{2nt^2}{1 - (n-1)/N}\right) \leq 2 \exp(-2nt^2).$$

C.2 The two-stage error bound

Combining the committee draw (Serfling) with the votes given the committee (Hoeffding for the Poisson–Binomial) yields the unconditional single-round error bound. This replaces the informal “for a random committee we expect $\bar{p}_C \approx \bar{p}$ ” step of earlier drafts with a proof.

Theorem C.4 (Single-round error in the uniform hierarchical model). *Let an odd-sized committee C of size n be drawn uniformly from S_N without replacement. Conditional on C , let $A_i = \mathbf{1}\{V_i = D_\infty\}$ be independent Bernoulli variables with means p_i , and let the panel choose the strict majority of its binary ballots. If $\gamma = \bar{p} - 1/2 > 0$, then for $n < N$,*

$$\mathbb{P}(d \neq D_\infty) \leq \inf_{0 < t < \gamma} \left\{ e^{-2n(\gamma-t)^2} + e^{-2nt^2/(1-(n-1)/N)} \right\} \leq 2e^{-n\gamma^2/2}.$$

When $n = N$, the sharper direct bound is $\mathbb{P}(d \neq D_\infty) \leq e^{-2n\gamma^2}$.

Proof. Write $S_C = \sum_{i \in C} A_i$ and split on the event $A_t = \{\bar{p}_C \geq \bar{p} - t\}$ for $0 < t < \gamma$. On A_t , the committee's mean agreement probability is at least $1/2 + \gamma - t$, so conditionally on the committee, by (7):

$$\mathbb{P}(d \neq D_\infty \mid C) = \mathbb{P}\left(\frac{S_C}{n} \leq \frac{1}{2} \mid C\right) \leq \exp\left(-2n\left(\bar{p}_C - \frac{1}{2}\right)^2\right) \leq e^{-2n(\gamma-t)^2}.$$

For $n < N$, the lower-tail form of Theorem C.2 gives

$$\mathbb{P}(A_t^c) \leq e^{-2nt^2/(1-(n-1)/N)}.$$

Add the two terms and optimize over t . Setting $t = \gamma/2$ and discarding the finite-population improvement yields the final coarse bound. When $n = N$, $\bar{p}_C = \bar{p}$ deterministically, so (7) applied at distance γ gives the direct claim. $\square$

Remark C.5 (Sharper at the top of the ladder). The optimized displayed bound retains the committee-selection improvement that the convenient $2e^{-n\gamma^2/2}$ cap discards. A deployed capacity successor is not automatically the $n = N$ case: its realized committee is produced by the protocol's union or active-set construction, under stake-weighted eligibility and possible identity history.

C.3 Successive probability-proportional-to-weight sampling

Let a fixed eligible set contain N validators with positive normalized weights w_i and values $p_i \in [0, 1]$. Successive sampling draws an ordered committee $(I_1, \dots, I_n)$ with

$$\mathbb{P}(I_j = i \mid I_1, \dots, I_{j-1}) = \frac{w_i}{1 - \sum_{\ell < j} w_{I_\ell}} \quad \text{for unselected } i.$$

Write $\pi_i^{(n)} = \mathbb{P}(i \in C)$ and define the design mean

$$m_{w,n} = \mathbb{E}\left[\frac{1}{n} \sum_{j=1}^n p_{I_j}\right] = \frac{1}{n} \sum_{i=1}^N \pi_i^{(n)} p_i, \quad \gamma_{w,n} = m_{w,n} - \frac{1}{2}.$$

The inclusion probabilities $\pi_i^{(n)}$ are generally neither nw_i nor a renormalized copy of the raw weights [7].

Theorem C.6 (Successive-PPSWOR panel error). *Let $0 < \alpha := \min_i w_i / \max_i w_i < 1$ and $\Delta := \max_i p_i - \min_i p_i$. Set*

$$v_{w,n} = \min \left\{ 4\Delta^2 n, \frac{1 + 4\alpha}{\alpha(1 - \alpha)} \Delta^2 N \left(\frac{N - n}{N} \right)^\alpha \right\}.$$

Conditional on the successive sample, let ballots be independent Bernoulli variables with means (p_{I_j}) . If $n < N$, $\gamma_{w,n} > 0$, and $\Delta > 0$, then for every $0 < t < \gamma_{w,n}$,

$$\mathbb{P}(d \neq D_\infty) \leq \exp\left(-\frac{n^2 t^2}{2v_{w,n}}\right) + \exp\left(-2n(\gamma_{w,n} - t)^2\right).$$

If $\Delta = 0$, committee selection is irrelevant and the direct bound is $e^{-2n\gamma_{w,n}^2}$. The same direct bound holds when $n = N$, because the committee is then the complete eligible set. For $\alpha = 1$, the design is uniform and Theorem C.4 applies instead.

Proof. For $X = \sum_{j=1}^n p_{I_j}$, Theorem 2 of Ben-Hamou, Peres, and Salez [3] gives

$$\mathbb{P}(X - \mathbb{E}X < -nt) \leq \exp\left(-\frac{n^2 t^2}{2v_{w,n}}\right).$$

Outside that lower-tail event, the sampled committee mean is at least $1/2 + \gamma_{w,n} - t$. Conditional Hoeffding in (7) supplies the second term; a union bound completes the proof. When $\Delta = 0$, X is deterministic. $\square$

This proof does not assume negative association. Such an argument would be invalid in general: successive weighted sampling can make two inclusion indicators positively correlated [1]. The price of arbitrary weight–value alignment is visible in α ; severe weight imbalance can make the bound vacuous. More importantly, the relevant margin is $\gamma_{w,n}$, which can vary with panel size even when weights and validator behavior are fixed.

Remark C.7 (Validity need not imply useful weighted-panel power). Theorem C.6 is a worst-case concentration statement, not a promise that its numerical upper bound is informative. For example, with $N = 1543$, $\Delta = 1$, and $\gamma_{w,n} = 0.10$, optimizing over t and clipping the result at one gives

n	$\alpha = 0.9$	$\alpha = 0.5$	$\alpha = 0.1$
95	1.000	1.000	1.000
383	0.981	0.981	0.981
767	0.791	0.791	0.791
1535	0.007	0.045	0.453

Thus the theorem supplies a valid transfer route while remaining nearly vacuous across much of the nominal ladder. The identical early columns occur because the $4\Delta^2 n$ branch of $v_{w,n}$ is active there; this branch need not remain active at the largest panel.

A Monte Carlo study of the selector alone can estimate the inclusion probabilities $\pi_i^{(n)}$, but not the design margin $\gamma_{w,n}$ without task-specific validator probabilities p_i . A deployment power study must therefore predeclare or estimate a joint stake–performance model, rather than infer statistical security from the stake vector alone.

Conditional on a fixed eligible set, fixed positive weights, and exclusions already determined, the pinned selector’s repeated weighted draws with duplicate rejection have the successive law above. This identifies a genuine one-round transfer route. It does not cover biased or ground beacon seeds, shared ballot state, a leader-specific ballot law, mid-round replacements, or a multi-round path unless the theorem’s parameters admit history-uniform bounds after every such update.

Remark C.8 (Dependence across rounds). Successive model committees may be dependent through the consumed set. For multi-round bounds we require, conditional on every reachable model history, that the next draw satisfy either Theorem C.4 or Theorem C.6, that its ballots satisfy the stated conditional-independence model, and that the applicable design margin and weight-ratio parameters obey history-uniform bounds. The resulting single-round cap can then be inserted into later path arguments without cross-round independence. We continue to write γ for a uniform-model lower margin below. Proving the required history-uniform premises for the full deployed reuse and replacement process remains open.

Remark C.9 (Successor unions are not fresh PPSWOR panels). If eligibility and weights remain fixed, drawing an initial committee and then drawing an appeal jury from its complement extends one ordered successive-sampling prefix. The active normal successor is nevertheless not that prefix: the construction removes the previous leader, and later paths can accumulate further history-dependent removals. Under unequal weights, deleting a selected element does not preserve the law of a fresh PPSWOR sample of the smaller size. Theorem C.6 therefore does not apply directly to the assembled successor.

There is a quantitative transfer route. If d values are removed from a successive prefix of length $n + d$ and all p_i lie in an interval of width Δ , the mean of the remaining n values differs from the prefix mean by at most $d\Delta/(n + d)$. A deletion-robust result can spend this amount of margin before applying the prefix concentration inequality. Establishing it for the deployed path still requires conditioning on leader selection, eligibility and quarantine changes, idle-seat replacements, and the event that the appeal path was reached.

D Optimistic Democracy as a G-AE-ADS Instance

The OD mapping uses Definition A.1: a normal proposing round creates the current column, an appeal jury is a REEVALUATE experiment on that frozen column, and a successful appeal returns to proposing so that a new leader can create a REGENERATE transition. Every evaluator round applies a strict-majority binary map ψ_r , and external transactions induce the action policy. Section H prices evaluator-seat work as $c_r = gn_r$.

Closed-form ladder calculations below use Definition A.2: they hold one column fixed and prescribe a fresh round kernel at every rung. The generative model and the fixed-column benchmark are now named separately; neither is asserted to be implementation-equivalent to the richer protocol state, outcome space, or terminal rule.

D.1 Committee Sizes: the Statistical Schedule

Definition D.1 (Committee Size Schedule). The statistical ladder analyzed in this paper is:

Normal rounds ($r = 0, 2, 4, \dots$) : 5, 11, 23, 47, $\dots$, 1535 (each $\approx 2 \times$ previous normal + 1)
 Appeal rounds ($r = 1, 3, 5, \dots$) : 7, 13, 25, 49, $\dots$ (previous normal + 2)

up to $R_{\max} = 18$ model rounds. When the required size exceeds the model population of 1535, the round uses that entire population (INVOLVE-ALL); in particular the uncapped request 1537 is represented by $n_r = 1535$. Let n_r denote the size at round r . The uncapped portion has geometric normal-round growth ($n_{r+2} \geq 2n_r$); after the first capacity hit, the model stays at the INVOLVE-ALL cap. Results that require continued geometric growth state an uncapped schedule separately.

Remark D.2. The pinned FeeManager exposes the nominal sizes 5, 7, 11, 13, $\dots$, 1535, 1537, but the realized protocol path is not this fixed interleaving. Appeal feasibility is checked against the round index and the currently available unconsumed validators, and the companion executable specification subtracts two seats after consecutive unsuccessful appeals. For the documented $N = 1543$ path, eight appeal juries through size 769 can lead to a 1535-member normal successor; this is a capacity scenario, not a protocol-wide appeal constant. Definition D.1 is the capped geometric abstraction used by this paper; Corollary A.14 separately removes the repeated cap when accounting for work through the first capacity-size round.

D.2 State Space

Definition D.3 (State Space). The state after round r is:

$$X_r = (r, P_r, k_r, \phi_r)$$

where P_r is the current frozen column, $k_r \in \{0, \dots, n_r\}$ is the Accept-vote count of round r (ballots are not accumulated), and $\phi_r \in \{\text{Active}, \text{Finalized}\}$. The committee size is determined by the model rung and d_r is a deterministic function of (n_r, k_r) . The fixed-column benchmark suppresses the constant coordinate P_r . Eligibility, replacement, and other protocol variables must be appended whenever they affect the next kernel.

D.3 Decision Rule

Definition D.4 (Quorum Decision). After round r with n_r validators and k_r Accept votes:

$$d_r = \begin{cases} \text{Accept} & \text{if } k_r > n_r/2 \\ \text{Reject} & \text{if } k_r \leq n_r/2 \end{cases}$$

Remark D.5 (Votes Are Not Accumulated). Each round r makes a decision based solely on the fresh ballots cast in that round. Ballots from previous rounds are *not* accumulated. Validator identities may nevertheless be reused in a normal successor.

Remark D.6. This paper uses the binary Accept-versus-non-Accept projection with aggregation threshold $1/2$. Part 1 analyzes any declared threshold and does not claim that $1/2$, or the global rule itself, is optimal. The protocol resolves strict-majority outcome categories and retains no majority as a distinct outcome for recovery and appeal comparison.

D.4 Appeal Mechanism

Definition D.7 (Appeal). After round r with decision d_r :

- Any agent may appeal by posting the required bond
- If appealed: purchase the next protocol experiment; an appeal jury re-evaluates P_r , while a successful challenge can lead to a later generated column $P_{r'} \neq P_r$
- If not appealed: $\phi_r \leftarrow \text{Finalized}$, system terminates with decision d_r

Definition D.8 (Tally-only appeal policy). A tally-only appeal policy is a reduced-form function

$$\pi : (n_r, k_r, d_r) \mapsto [0, 1]$$

giving the probability that someone appeals given the current public tally. The general AE-ADS policy may instead depend on the full history H_r , including private signals when the policy models a particular watcher.

Remark D.9. The appeal policy is external to the protocol. It depends on:

- Economic incentives (stake required vs potential gain)
- Information available to potential appellants
- Beliefs about whether the decision matches a desired reference

D.5 Termination

Definition D.10 (Maximum Appeals). The statistical model contains $R_{\max} = 18$ rounds indexed $r = 0, \dots, R_{\max} - 1$. The last modeled round finalizes regardless of the appeal policy.

Remark D.11 (The Top of the Ladder). In the statistical ladder, an INVOLVE-ALL round has $\bar{p}_C = \bar{p}$ exactly. In the protocol, capacity exhaustion instead changes whether an appeal can be opened and how a successor committee is constructed. At archived revision 8795606e, also pinned by the companion TLA repository, the voting rule compares the winning category with half the *realized* committee length. A 1535-member successor therefore has strict-majority threshold 768, not a frozen threshold 772 inherited from an active set of size 1543. Section J quantifies the statistical ladder, not this path-dependent capacity construction.

Definition D.12 (Stopping Time).

$$\tau = \min\{r : \text{no appeal after round } r\} \wedge (R_{\max} - 1)$$

D.6 The Verified State Machine, and What This Model Abstracts

The chain of Definition D.3 is an abstraction of the verified protocol state machine (Part 3): a round bundles the leader-proposal, commit, and reveal phases into a single vote draw, and projects the vote-complete *decided statuses* (Accepted, Undetermined, and timeout statuses) into the binary d_r . This projection collapses the protocol's strict-majority outcome categories and distinct no-majority outcome. The following deployed behaviors are *not* in the idealized model:

1. **Changing resolutions.** An appeal jury re-evaluates a challenged proposal, but a normal successor re-enters Proposing with a new leader and execution result. Definition A.1 represents this as a column transition; the later closed-form ladder bounds use the fixed-column specialization and therefore do not estimate that transition kernel.
2. **Path-dependent sizes and capacity.** Consecutive unsuccessful appeals use the companion specification's minus-two adjustment, while contract appeal feasibility depends on available unconsumed validators. Round parity alone therefore does not determine the next realized size or the number of appeals.
3. **Distinguished and adaptive seats.** A normal-round leader belongs to the voting array, and idle validators can be replaced during commit. The realized committee need not be the initially sampled committee, and the leader's ballot need not have the same law as an ordinary evaluator's.
4. **Voided appeal rounds.** If an appeal round hangs (all or most of its committee idle), the implementation's idleness path *voids* the round: non-committed validators are slashed, the transaction reverts to its pre-appeal status, and a reduced appeal window opens. This path changes cost and round-budget accounting and is outside the chain analyzed below.
5. **The outcome race.** The same hanging appeal, detected through a different code path, can instead yield Undetermined, a divergence tracked as `genlayer-consensus#1157`. A binary rejection is not equivalent to restoring the challenged status, so no result in this paper is asserted for both branches without an explicit refinement proof.
6. **Clock time.** Appeal windows, explicit finalization calls, and phase timeouts are not represented by the round-index latency penalty $\lambda\tau$ alone.

E Connection to Part 1

E.1 Each OD round as aggregate disambiguation

Proposition E.1 (Binary OD projection instantiates the Part 1 rule). *At round r , the decision d_r equals Part 1's inclusive threshold decision at threshold $1/2$ over the round- r committee, because every modeled n_r is odd:*

$$d_r = \mathbf{1} \left[\frac{k_r}{n_r} > \frac{1}{2} \right]$$

This matches the protocol's Accept-versus-non-Accept projection: acceptance requires a strict majority of agree votes, while the full protocol retains the non-Accept outcome categories for recovery and appeal comparison.

Remark E.2 (The deployed leader is a distinguished voter). The archived normal-round commit path collects ballots across the full committee, including its leader, and the pinned voting rule counts the complete ballot array. The homogeneous table below nevertheless models five exchangeable independent ballots. If leader behavior is mechanically or strategically tied to the proposal, its coordinate must instead be modeled separately (for example as a distinguished Bernoulli or deterministic coordinate in a Poisson–Binomial count). The symmetric round-zero numbers are therefore model benchmarks, not probabilities for the deployed leader–validator interaction.

The proposition is deliberately round-local. The stopped output d_r also depends on the history-dependent policy and kernels, so its guarantees come from the AE-ADS analysis rather than from treating the whole execution as one fixed panel covered automatically by Part 1.

E.2 Single-Round Error Probability

Definition E.3 (Round- r Error).

$$\varepsilon_r = \mathbb{P}(d_r \neq D_\infty(P_r))$$

where the probability is over both the committee draw and the votes.

Theorem E.4 (Error Probability Bound). *For every statistical-model round r satisfying the conditional assumptions of Remark C.8:*

$$\varepsilon_r \leq 2 \exp\left(-\frac{n_r \gamma^2}{2}\right).$$

Proof. This is the coarse form of Theorem C.4 applied conditionally on the model history; the Serfling term controls the uniform committee-mean deviation and Hoeffding controls the ballots given that committee. $\square$

Remark E.5 (Weighted analogue). For a successive weighted round, replace the displayed cap by the optimized right-hand side of Theorem C.6, using its panel-size- dependent design margin γ_{w,n_r} and weight ratio α_r . This is a one-round theorem, not yet a certificate for a successor assembled through reuse, replacement, or a distinguished leader ballot.

E.3 Numerical Examples: Exact, Not Bounds

At $n_0 = 5$ the exponential bound is *vacuous* ($2e^{-5\gamma^2/2} \geq 1$ for all $\gamma \leq 1/2$); it becomes informative once $n_r > 2 \ln(2)/\gamma^2$. For small committees the honest tool is the exact distribution. In the homogeneous approximation ($p_i \equiv \bar{p}$; the Poisson–Binomial refines this), the round-0 error is $\mathbb{P}(\text{Bin}(5, \bar{p}) \leq 2)$:

Agreement margin γ	$\bar{p}$	Exact ε_0 ($n_0=5$)	Bound informative from $n \geq$
0.10	0.60	31.7%	139
0.20	0.70	16.3%	35
0.30	0.80	5.8%	16
0.40	0.90	0.86%	9
0.45	0.95	0.12%	7

Proposition E.6 (Self-consistent leader sensitivity benchmark). *Suppose the five-seat normal panel contains one leader who deterministically votes Agree with its own proposal, while the other four seats vote Agree independently with probability p_A . If $X \sim \text{Binomial}(4, p_A)$, the panel accepts exactly when $X \geq 2$. Its error is therefore*

$$\begin{aligned} \mathbb{P}(X \leq 1) & \text{ when the population target accepts,} \\ \mathbb{P}(X \geq 2) & \text{ when the population target rejects.} \end{aligned}$$

In particular, the following comparison holds:

population target	p_A	five exchangeable seats	four seats + fixed Agree leader
Accept	0.70	16.3%	8.4%
Reject	0.30	16.3%	34.8%
Reject	0.20	5.8%	18.1%

Proof. The fixed leader contributes one Agree vote, and strict majority of five requires three Agree votes. The displayed binomial tails and numerical values follow immediately. $\square$

The proposition is deliberately conditional: the contract includes the leader’s seat but does not mechanically force its ballot hash to equal its proposal hash. It quantifies the asymmetric effect of a self-consistent leader without presenting that behavior as the deployed ballot law.

Remark E.7. With only 5 validators, population-disagreement probability is material except at large homogeneous agreement margins. These numbers are oracle appeal-rate benchmarks only: a perfect oracle appeals exactly on those events. A real appeal rate additionally depends on beliefs, private signals, transaction utility, bonds, and protocol transitions, so it cannot be read off this table.

F Convergence Under Appeals

We now analyze what happens when errors are appealed in the fixed-column, fresh-round specialization. Thus D_∞ is constant throughout this section. The changing-column G-AE-ADS instead requires the conditional certificates or transition controls of Section A.

F.1 Appeal Policy: Always Appeal on Error

Definition F.1 (Perfect Appeal Policy). The perfect appeal policy is:

$$\pi^*(n_r, k_r, d_r) = \begin{cases} 1 & \text{if } d_r \neq D_\infty \\ 0 & \text{if } d_r = D_\infty \end{cases}$$

Remark F.2. This is an unimplementable reference policy: it observes D_∞ , appeals every nonmatching decision, and never appeals a matching one. It is a model benchmark, not an attainable performance guarantee.

For later use, let

$$\bar{\varepsilon}_r = \min\left\{1, 2 \exp\left(-\frac{n_r \gamma^2}{2}\right)\right\}$$

denote the history-uniform conditional cap supplied by Theorem E.4 under Remark C.8. Define

$$r_0 = \min(\{r : n_r \gamma^2 > 2 \ln 2\} \cup \{R_{\max}\}).$$

Then every displayed cap before r_0 equals one, so the informative part of the perfect-policy product is exactly $\prod_{r=r_0}^{R_{\max}-1} \bar{\varepsilon}_r$, with the usual empty-product value one. The early factors are retained in formulas only to keep the path accounting uniform; they contribute no numerical contraction.

Theorem F.3 (Finite-Ladder Residual under Perfect Appeals). *Under the perfect appeal policy π^* , the finite ladder can finalize on a wrong decision only if every modeled round is wrong. Consequently,*

$$\mathbb{P}(d_\tau \neq D_\infty) \leq \prod_{r=0}^{R_{\max}-1} \bar{\varepsilon}_r.$$

In the homogeneous simulation model with independent fresh rounds, equality holds with the exact binomial errors ε_r in place of $\bar{\varepsilon}_r$.

Proof. At every nonterminal round the oracle appeals a wrong decision and stops at a matching decision. The last modeled round cannot be appealed, so residual error is the intersection of all round-error events, not the empty event. Iterated conditioning and the history-uniform caps give the product bound. Independence gives the stated equality in the homogeneous model. $\square$

F.2 Appeal Policy: Appeal with Probability q on Error

Definition F.4 (Imperfect Appeal Policy). An imperfect appeal policy appeals with probability $q \in (0, 1]$ when the decision is wrong:

$$\pi_q(n_r, k_r, d_r) = \begin{cases} q & \text{if } d_r \neq D_\infty \\ 0 & \text{if } d_r = D_\infty \end{cases}$$

Theorem F.5 (Finite-Ladder Error under Imperfect Oracle Detection). *Suppose that, whenever a nonterminal round is wrong, an oracle detector appeals with history-independent probability $q \in (0, 1]$, and never appeals a matching decision. Then*

$$\mathbb{P}(d_\tau \neq D_\infty) \leq \sum_{r=0}^{R_{\max}-2} (1-q)q^r \prod_{s=0}^r \bar{\varepsilon}_s + q^{R_{\max}-1} \prod_{s=0}^{R_{\max}-1} \bar{\varepsilon}_s.$$

Proof. To finalize incorrectly at a nonterminal round r , all earlier rounds must be wrong and appealed, round r must be wrong, and its appeal must be missed. The last term is the event that every earlier wrong round is appealed and the terminal round is also wrong. These terminal events are disjoint; iterated conditioning gives the displayed bound. $\square$

Corollary F.6. *If $q = 1$, the residual is bounded by $\prod_{r=0}^{R_{\max}-1} \bar{\varepsilon}_r$; finite depth does not make it identically zero.*

F.3 Margin-Responsive Heuristic Policy

An implementable tally-only heuristic can appeal with probability

$$\pi_{\text{margin}}(n_r, k_r, d_r) = f\left(\left\lceil \frac{k_r}{n_r} - \frac{1}{2} \right\rceil\right)$$

for a decreasing $f : [0, 1/2] \rightarrow [0, 1]$. This makes close observed tallies more likely to be appealed by construction. Whether that behavior is rational depends on the conditional appeal-success probability and utility model. The observed margin alone does not identify either quantity.

G OD as a Family of Markov Chains

Fix the column and assume the fresh-round abstraction: conditional on r , the next count has a prescribed law Q_r independent of earlier counts. After suppressing the constant P_r , the finite state $S = \{(r, k, \phi)\}$ is Markov. A reduced policy $\pi : S \rightarrow [0, 1]$ and the count law induce

$$\begin{aligned} P_\pi((r, k, \text{Active}), (r+1, j, \text{Active})) &= \pi(r, k) Q_{r+1}(j), \\ P_\pi((r, k, \text{Active}), (r, k, \text{Finalized})) &= 1 - \pi(r, k), \end{aligned}$$

with forced absorption at the terminal round. In the homogeneous numerical model, $Q_r = \text{Binomial}(n_r, 1/2 + \gamma)$. The null policy is $\pi_0 \equiv 0$; the oracle policy π^* was defined in the preceding section. Expected stopping time, error, and cost are properties of the pair (Q, π) , not of the size ladder alone.

This reduction is unavailable for the full G-AE-ADS unless the current column and any variables governing generator, eligibility, reuse, replacement, and appeal status are added to the state. In particular, writing (r, k, ϕ) after a successful appeal would discard precisely the horizontal matrix move introduced in Proposition A.9.

Proposition G.1 (Stopping and Cost Monotonicity). *If $\pi \preceq \pi'$, then:*

1. $\mathbb{E}_\pi[\tau] \leq \mathbb{E}_{\pi'}[\tau]$
2. $\mathbb{E}_\pi[G] \leq \mathbb{E}_{\pi'}[G]$

under the natural coupling that uses the same round draws and appeal uniforms.

Proof. Pointwise policy ordering gives pathwise stopping and cost ordering. It gives no general ordering of final error: an additional appeal can replace a matching decision by a nonmatching one. $\square$

Remark G.2 (No automatic consistency monotonicity). Error improves under policies that selectively continue after nonmatching decisions, such as the oracle benchmark. For implementable policies this is an assumption or a theorem to prove, not a consequence of making more appeals.

H Evaluator-Work Cost in the OD Instance

Definition A.3 allows arbitrary real resource, latency, and error costs. This section isolates one component: evaluator-seat work in the statistical OD model. It is not a complete welfare calculation and does not count bonds returned between participants as social cost.

H.1 Cost Definition

Definition H.1 (Round Cost). The cost of round r is proportional to the committee size:

$$G_r = g \cdot n_r$$

where g is the per-validator cost (gas).

Definition H.2 (Total Cost).

$$G = \sum_{r=0}^{\tau} G_r = g \sum_{r=0}^{\tau} n_r$$

H.2 Expected Cost Under Different Policies

Proposition H.3 (No Appeals). *If no one ever appeals ($\pi \equiv 0$):*

$$\mathbb{E}[G] = g \cdot n_0 = 5g$$

Proposition H.4 (Perfect Appeals). *Under the perfect appeal policy π^* in the independent fresh-round model:*

$$\mathbb{E}[G] = g \cdot n_0 + \sum_{r=1}^{R_{\max}-1} g \cdot n_r \cdot \mathbb{P}(\text{reach round } r)$$

where $\mathbb{P}(\text{reach round } r) = \prod_{s=0}^{r-1} \varepsilon_s$.

Proof. Under π^* , round r is reached iff all previous independent rounds had errors:

$$\mathbb{P}(\text{reach round } r) = \prod_{s=0}^{r-1} \mathbb{P}(d_s \neq D_\infty) = \prod_{s=0}^{r-1} \varepsilon_s$$

□

H.3 Clear vs Ambiguous Proposals

Remark H.5 (Agreement margin does not determine cost by itself). For the oracle benchmark, high agreement margins make an early matching round likely and keep expected cost near $5g$. At small margins, each oracle-tested round is wrong with probability near $1/2$, so expected depth increases but the system still does not use every round with high probability. A policy that always appeals uses the whole ladder at every margin, while the null policy always costs $5g$. Workload agreement margins and appeal behavior must therefore be modeled jointly.

H.4 Voided Rounds

Remark H.6 (Cost of voided appeal rounds). Under the implemented idleness semantics (Section D.6), an appeal round may be *voided*: work is performed but the transaction state returns to its pre-appeal status. This is absent from the Markov and cost models above. A geometric inflation factor $1/(1 - \nu)$ would be valid only under an additional history-independent retry model with void probability ν and no binding retry cap; we do not assume it here. The fee classification remains an open protocol decision (Part 3, Finding F3).

I Comparison with Part 1

	Part 1 (Fixed K)	Part 2 (G-AE-ADS / idealized OD)
Sampling object	declared generator–evaluator population	adaptive columns, panels, and action policy
One-panel law	hypergeometric or binomial	uniform or successive-PPSWOR draw plus conditional Poisson–Binomial
Primary target	population/census consistency	consistency for the column finalized at policy-dependent stopping
Panel size	declared K	$n_0 = 5$, then model schedule if appealed
Economic object	outside scope	resource, latency, and error-loss objective
Stopping/action	fixed design	Bellman planner over stop, re-evaluate, or regenerate
Certification	exact tails and population-level certificates	generic adaptive composition theorem; no instantiated live OD certificate

Part 1 supplies no baseline $K = 1000$. Corollary A.14 proves that the nominal schedule has bounded arithmetic overhead relative to an oracle-known adequate panel, but it does not prove that deployed OD is cheaper on a given workload. Such a comparison still requires a declared workload law, a target error budget or error loss, an implementable appeal policy, and comparable sampling assumptions.

J The Ladder: Design Bounds and Appeal Rates

Two quantitative diagnostics of the statistical schedule are design-relevant enough to state on their own. Neither is a deployment certificate without the population and sampling-law requirements of Part 1.

J.1 A Sufficient Top-Round Design Margin

Proposition J.1 (Top-Round Hoeffding Design Margin). *In the homogeneous i.i.d. benchmark, or in an all-population model round whose independent agreement indicators have average $1/2 + \gamma$, the sufficient condition*

$$\gamma \geq \sqrt{\frac{\ln(1/\delta)}{2n_{\text{top}}}}.$$

ensures that the top round's disagreement probability is at most δ .

Proof. Apply Hoeffding's upper bound $\mathbb{P}(d_{\text{top}} \neq D_{\infty}) \leq \exp(-2n_{\text{top}}\gamma^2)$ and solve for γ . $\square$

Example J.2. With $n_{\text{top}} = 1535$ and $\delta = 10^{-6}$, the sufficient value is $\sqrt{\ln(10^6)/3070} \approx 0.0671$. Exact binomial inversion in the homogeneous benchmark gives the smaller threshold 0.06043. The difference is ordinary Hoeffding conservatism. Below either reported number, failure of that particular test is inconclusive: it is not a proof that no appeal policy, no alternative rule, or no larger panel can meet the target.

Part 1's operational certificate answers a different question. A predeclared calibration matrix indexed by generators and evaluators can lower-bound the mass of future resolutions for which an independent K -panel reproduces the declared population decision. Reusing one live round's vote margin as both estimator and decision does not instantiate that theorem. A protocol-level *contentious* outcome remains a plausible design feature, but it requires a separately specified, selection-valid calibration rule satisfying Theorem A.10.

J.2 Round-Zero Error and Appeal Pressure

The round-0 errors of Section E.3 are oracle benchmarks, not appeal rates. A perfect oracle appeals exactly on those events. A fresh watcher with signal agreement probability p_a , independent of the round conditional on D_{∞} , appeals whenever its signal disagrees with the round, with probability

$$q_A = (1 - p_a)(1 - \varepsilon_0) + p_a \varepsilon_0.$$

Other policies have different rates.

Agreement margin γ	Round-0 error	Perfect-oracle benchmark
0.30–0.45	0.1%–6%	oracle appeals are rare
0.20	~16%	oracle appeals about one round in six
0.10	~32%	oracle appeals are common

The expected ladder cost per transaction is policy-composable from Section H. Part 4 may use these values only after naming the appeal policy; treating round error as observed appeal frequency is not valid without the perfect-oracle assumption.

K Decentralized Stopping: The Appellant's Private Rule

Theorem A.5 says when a social planner should purchase the next experiment. OD delegates that action to an appellant. We now model the appellant's private rule and make explicit that private profitability need not have the same sign as social continuation value. Part 1 supplies the reference decision and one-panel sampling laws; the finite-ladder outcome depends on the appeal policy and on the Part 2 transition model. Part 4 analyzes the implemented fee game and the resulting implementation gap in greater detail.

K.1 An All-Population Model-Round Bound

Theorem K.1 (All-Population Model-Round Bound). *Suppose one statistical-model round uses every member of an eligible set of size N' , its agreement indicators are independent, and their mean is $1/2 + \gamma_{\text{top}}$. Then committee-draw error vanishes and Hoeffding gives*

$$\mathbb{P}(d_{\text{top}} \neq D_{\infty}) \leq \exp(-2N' \gamma_{\text{top}}^2).$$

For example, $N' = 1000$ and $\gamma_{\text{top}} = 0.1$ give the sufficient bound $e^{-20} \approx 2 \times 10^{-9}$. We write ε_{top} for a valid bound of this kind.

Remark K.2. The bound is sufficient and parameter-dependent, not exact. It also does not apply verbatim to the deployed capacity endgame, whose eligible set can omit earlier leaders and whose realized committee is constructed from protocol history. The deployed vote threshold is strict majority of that realized committee, not a frozen-universe threshold.

K.2 The Appellant's Role

The appellant's role is to trigger escalation. Whether escalation improves population consistency depends on the appellant's information and the next round's law.

Definition K.3 (Appellant as Trigger). An appellant who believes the current decision is population-inconsistent can:

1. Post bond/cost c to appeal
2. Observe the outcome category of the next protocol round
3. If the appeal succeeds under the protocol's category-comparison rule: recover the bond and receive the specified return
4. If the appeal fails: forfeit the bond according to the fee rules

K.3 Appellant Model

Definition K.4 (Appellant). An appellant a has:

- **Private signal** $V_a \in \{0, 1\}$: their own execution (e.g., running the contract with their LLM)
- **Subjective agreement belief** $\hat{p}_a$: their belief about agreement with the declared population decision
- **Observed votes** (n_r, k_r) : the public vote count from round r

Remark K.5 (The Appellant Does Not Know Their True p_a). The appellant generally does not know $p_a = \mathbb{P}(V_a = D_{\infty})$. It must be estimated under a declared task/evaluator law or treated as a subjective belief. It is not semantic accuracy unless D_{∞} has separately been validated against external labels.

K.4 Bayesian Update Using Observed Votes

A Bayesian appellant may update a belief about D_{∞} using both its private signal and the observed votes, but only after specifying a prior over the competing population models and a likelihood for the committee sampler.

Definition K.6 (Posterior from a Declared Bayesian Model). Let Λ be a prior over D_{∞} and all nuisance parameters needed to specify the round likelihood. After observing k_r Accept votes out of n_r , define

$$b_r = \mathbb{P}(D_{\infty} = 1 \mid n_r, k_r)$$

where the probability is taken under Λ and the declared likelihood. The count (n_r, k_r) alone does not determine b_r .

Proposition K.7 (Bayesian Posterior). *Let b_r be the public posterior after the observed votes. After incorporating private signal V_a :*

$$\mathbb{P}(D_{\infty} = 1 \mid V_a, n_r, k_r) = \frac{b_r \mathbb{P}(V_a \mid D_{\infty} = 1)}{b_r \mathbb{P}(V_a \mid D_{\infty} = 1) + (1 - b_r) \mathbb{P}(V_a \mid D_{\infty} = 0)}.$$

Remark K.8 (Simplified Model). For tractability, the implementable watcher bound below uses a fixed signal agreement parameter p_a and a specified appeal rule. The plug-in margin score used in the numerical diagnostic is not called a Bayesian posterior.

K.5 Appeal Incentives

Definition K.9 (Payoff Structure). Let:

- c : cost to appeal (stake)
- b : net bonus received if the protocol classifies the appeal as successful
- d_r : current decision
- S : the event that the appeal's outcome category differs from the challenged category

Remark K.10 (Protocol success versus binary reversal). The deployed protocol compares full outcome categories; no majority and deterministic violation can therefore make an appeal successful. The binary flip model below is a projection and must not be read as the complete fee classification.

Remark K.11 (Distinct adversarial models). Part 1's α is a non-adaptive contamination fraction or, in a separate benchmark, a fixed Byzantine network share. An appeal budget is neither of those quantities. Relating stake ownership to the number of affordable appeals requires the bond schedule and fee flows; relating it to outcome manipulation additionally requires the deployed selection law. No conversion is asserted here.

Proposition K.12 (Expected Payoff from Appealing). *Fix the accounting relative to not appealing (net framing): a winning appeal nets $+b$, a losing appeal nets $-c$. If the appellant assigns probability s to the appeal succeeding, then*

$$\mathbb{E}[\text{payoff}] = s b - (1 - s) c,$$

which is positive iff

$$s > \frac{c}{b + c}.$$

Definition K.13 (Decentralized implementation criterion). At history h , define the planner's continuation advantage by

$$\Delta_{\text{soc}}(h) = s_r(h) - \mathbb{E}[a_{r+1}(h, H_{r+1}) + V_{r+1}(H_{r+1}) \mid h, \text{continue}].$$

Let $\Delta_i(h)$ be potential appellant i 's private expected net gain from appealing, including the bond payoff, transaction utility, and any private cost. If any willing actor can trigger the next round, we say decentralized appeals implement the planner rule when, at every reachable non-tie history,

$$\mathbf{1}\left[\max_i \Delta_i(h) > 0\right] = \mathbf{1}[\Delta_{\text{soc}}(h) > 0].$$

The left indicator records whether the decentralized mechanism continues; Theorem A.5 identifies the right indicator as the planner's optimal action.

Remark K.14 (Funding work is not incentive alignment). For the bond-only payoff of Proposition K.12, $\Delta_i(h) = s_i(h)b - (1 - s_i(h))c$. Protocol-defined appeal success is not the same event as reducing population-decision loss, and private transaction value need not equal social value. Consequently a bond can exactly pre-fund the next round while Definition K.13 fails. Part 4 studies the implemented payoffs; calibrated beliefs and a multi-appellant equilibrium are still required to quantify the gap.

Remark K.15 (Implemented parameters). The deployed protocol returns $1.5\times$ the bond on a successful appeal (Part 3; bond posting and resolution), i.e. $b = c/2$, giving the success threshold

$$s > \frac{c}{c/2 + c} = \frac{2}{3}.$$

Under this schedule, appealing is rational only when the appellant believes the flip is more likely than not by a wide margin — a deliberately conservative design. Earlier drafts computed the threshold as $c/(b + 2c)$ by double-counting the returned stake as a gain; the net framing above is the correct accounting.

K.6 The Margin-Based Appellant and the Bond Schedule

What success probability s should a rational appellant assign? The public vote count is relevant, but Part 1 does not turn one realized margin into that posterior. Its operational certificate uses a separate, predeclared generator-evaluator matrix and certifies future panels.

Definition K.16 (Uncalibrated Plug-In Flip Score). Let $\hat{\mu}_r = k_r/n_r$ and $\hat{\gamma}_r = |\hat{\mu}_r - 1/2|$. The numerical study reports the plug-in score

$$h(\hat{\gamma}_r, n_{r+1}) = \Phi\left(-\frac{\hat{\gamma}_r \sqrt{n_{r+1}}}{\hat{\sigma}_r}\right), \quad \hat{\sigma}_r^2 = \hat{\mu}_r(1 - \hat{\mu}_r).$$

It is the Gaussian next-round tail after substituting the current empirical mean for an unknown homogeneous mean. It is not a calibrated probability, a confidence bound, or a Part 1 certificate. In particular, fresh rounds do not use Part 1's nested-panel bridge law.

Example K.17 (The asymmetry the flat bond leaves open). After a 3–2 round-0 vote, the score is about 30%. The actual next-round flip probability cannot be recovered from that count alone; in the homogeneous sensitivity study it varies sharply with the true margin and with the direction of the current decision. The score is therefore useful as a diagnostic feature, not as a priced probability.

Proposition K.18 (Break-Even Bond Identity). *If s is a calibrated appeal-success probability and the protocol wants break-even at $s = s^*$, then the bonus-to-bond ratio must satisfy*

$$\frac{b}{c} = \frac{1 - s^*}{s^*},$$

and, with fixed b , $c = bs/(1 - s)$ makes an appellant with success belief s exactly indifferent.

Remark K.19 (Design recommendation). The identity motivates studying margin-dependent bonds, but it does not by itself supply the calibrated s . Any implementation must estimate appeal success under the full outcome-category rule, account for selection and post-selection, and analyze strategic manipulation of the public vote count. The plug-in score above is not sufficient for deployment.

K.7 The Signal-Based Appellant

The margin-based model uses public information only. An appellant with a *private signal* (their own execution) combines both:

Corollary K.20 (Rational Appeal Condition, Signal-Based). *Under risk-neutral expected utility, an appellant appeals exactly when its posterior probability of protocol-defined success exceeds $c/(b + c)$. The posterior must come from a declared belief model; neither the plug-in score nor the private signal alone determines it without further assumptions.*

Remark K.21 (True vs. Believed Agreement). The condition uses a subjective success belief, not the true agreement probability p_a . Whether repeated losses correct miscalibration is a learning/equilibrium question and is not proved here.

Example K.22 (Numerical Thresholds, net framing). $b = c$: threshold 1/2. $b = 2c$: threshold 1/3. Implemented $b = c/2$: threshold 2/3.

K.8 When Appellants Don't Appeal

Proposition K.23 (No Appeal Conditions). *For the risk-neutral bond-only payoff of Proposition K.12, an appellant does not appeal when its calibrated posterior protocol-success probability is at most $c/(b + c)$. Agreement of its private signal with the current decision may affect that posterior, but is neither necessary nor sufficient once transaction utility or other information is included.*

K.9 Finite-Ladder Error with a Watcher

The finite-ladder bound requires a signal-dependence assumption that earlier drafts left implicit:

Definition K.24 (Fresh-Signal Appellant). A *fresh-signal watcher* re-executes the task at every round. We assume its signals $V_a^{(0)}, V_a^{(1)}, \dots$ are conditionally independent across rounds and independent of the corresponding committee ballots given D_∞ , with $\mathbb{P}(V_a^{(r)} = D_\infty) = p_a$. A product law for all private inputs can justify this assumption; merely choosing distinct seeds cannot exclude shared random external state.

Without re-execution, a single signal is drawn *once*: if it is wrong (probability $1 - p_a$), it stays wrong at every round, and no exponent in $R_{\max}$ is available.

Theorem K.25 (Finite-Ladder Bound with a Fresh-Signal Watcher). *Suppose a watcher satisfying Definition K.24 appeals every nonterminal round for which its signal differs from the current binary decision. Then*

$$\mathbb{P}(d_\tau \neq D_\infty) \leq (1 - p_a) \sum_{r=0}^{R_{\max}-2} \bar{\epsilon}_r + \bar{\epsilon}_{R_{\max}-1}.$$

Proof. At a nonterminal round, wrong finalization requires both a wrong round decision and a watcher signal that agrees with that wrong decision. Conditional independence bounds this by $(1 - p_a)\bar{\epsilon}_r$. These finalization events are disjoint, so summing them and adding the terminal-round cap proves the result. The bound is deliberately conservative because it discards the probability of reaching each round. $\square$

Remark K.26 (True vs. Subjective Agreement). The bound depends on the true reference-agreement probability p_a , not the watcher's stated confidence. It does not claim semantic correctness or an exponent in $R_{\max}$ from one persistent signal.

Corollary K.27 (Numerical Example). *In the homogeneous independent-round simulation with $\gamma = 0.2$ and $p_a = 0.9$, the measured end-to-end error is about 2%, compared with a 16.3% round-0 error. This is a model diagnostic, not a deployment estimate. With m conditionally independent watchers, the nonterminal miss factor for a wrong decision becomes $(1 - p_a)^m$, while spurious appeals after matching decisions also become more likely.*

The watcher need not observe D_∞ , but the bound requires a declared reference-agreement probability, the conditional independence in Definition K.24, and an appeal rule that actually triggers on signal disagreement. Part 1 does not certify watcher participation, beliefs, or the finite protocol ladder.

K.10 Adversarial Appellants

Proposition K.28 (An Extra Appeal Can Reverse a Match). *Suppose a statistical-model round matches D_∞ and is appealed. The next round disagrees with D_∞ with its own conditional error probability, bounded in the uniform model by $\bar{\epsilon}_{r+1}$. Thus an extra appeal can replace a matching decision by a nonmatching one.*

The bond prices the attempt and the protocol bounds the number of appeal opportunities. Those economic facts may bound grieving cost, but they do not imply statistical outcome immunity. Any adversarial-outcome claim must combine the appeal policy, finite depth, full outcome categories, committee-selection law, and Part 1's separate population-contamination model.

K.11 Multi-appellant boundary

With several potential appellants, a public-good incentive to wait competes with transaction-ordering incentives to preempt rivals. Which force dominates depends on mempool visibility, exclusive state changes, failed-transaction gas, private values, and the category-level success rule. None follows from the sampling theory. Part 4 analyzes a reduced simultaneous volunteer game; the ordered preemption game remains a deployment-level mechanism problem.

L Oracle Cost Scaling and Sequential-Testing Comparison

Three comparisons must remain separate. Theorem A.5 characterizes the optimal policy after a workload, likelihood, and social loss have been declared. Corollary A.14 shows that the nominal ladder has bounded schedule overhead if an oracle reveals the adequate rung. We now compare an oracle-aided stochastic cost benchmark with classical sequential testing. This last comparison is informative about scale, but it cannot establish statistical optimality: the oracle policy observes D_∞ , whereas a test is trying to infer which hypothesis holds.

L.1 The Wald Benchmark

Theorem L.1 (Wald [5, 6]). *Consider testing $H_0 : p = p_0$ vs $H_1 : p = p_1$ with error constraints α, β . Among all sequential tests satisfying these constraints, the Sequential Probability Ratio Test (SPRT) minimizes the expected sample size $\mathbb{E}_{p_0}[n]$ and $\mathbb{E}_{p_1}[n]$.*

Remark L.2. SPRT achieves:

$$\mathbb{E}_p[n] = \Theta\left(\frac{1}{D(p \| 1/2)}\right) = \Theta\left(\frac{1}{\gamma^2}\right)$$

where $\gamma = |p - 1/2|$ and $D(\cdot \| \cdot)$ is the KL divergence.

L.2 OD vs Classical Sequential Testing

OD with rational appellants differs from SPRT:

Aspect	SPRT	OD + Rational Appellants
Stopping rule	Likelihood ratio threshold	No one appeals
Who decides	Protocol (automatic)	Agents (strategic)
Information	Votes only	Votes + private signals
Error control	Test-design guarantee	Depends on appeal policy and round laws

L.3 Oracle Benchmark Upper Scaling

The next result is only a cost upper bound for the ideal benchmark.

Theorem L.3 (Oracle Cost Upper Bound). *In the independent fresh-round model, suppose the uncapped schedule grows geometrically in the sense that $c_1 \rho^r \leq n_r \leq c_2 \rho^r$ for constants $\rho > 1$ and $c_1, c_2 > 0$, and suppose the modeled ladder reaches a size of order $1/\gamma^2$. Under π^* ,*

$$\mathbb{E}[n_\tau^{\text{total}}] = \mathbb{E}\left[\sum_{r \leq \tau} n_r\right] \leq \frac{C}{\gamma^2}$$

for a constant C depending only on the schedule constants.

Proof. Under π^* , round r is reached only if round $r - 1$ is wrong. Hence

$$\mathbb{P}(\text{reach } r) \leq \min\{1, 2e^{-n_{r-1}\gamma^2/2}\}.$$

Let $r^* = \min\{r : n_r \gamma^2 \geq 2\}$. Geometric growth gives $n_{r^*} = O(1/\gamma^2)$ and

$$\begin{aligned} \mathbb{E}[n_\tau^{\text{total}}] &\leq \sum_{r \leq r^*} n_r \\ &\quad \leq c' n_{r^*} \\ &\quad + \sum_{r > r^*} n_r \cdot 2e^{-n_{r-1}\gamma^2/2}. \end{aligned}$$

The first sum is geometric and therefore $O(n_{r^*})$. In the second, n_r/n_{r-1} is bounded and n_{r-1} grows geometrically, so the exponential tail is summable with first term $O(n_{r^*})$. Both terms are $O(1/\gamma^2)$. $\square$

Proposition L.4 (Sequential Lower Bound). *Any vote-only decision procedure that does not observe the true hypothesis and distinguishes i.i.d. Bernoulli populations with $p = 1/2 + \gamma$ from $p = 1/2 - \gamma$ with both error probabilities at most $\delta < 1/2$ satisfies the information lower bound [5, 6]:*

$$\mathbb{E}[n] \geq \frac{(1 - 2\delta) \ln \frac{1-\delta}{\delta}}{D\left(\text{Bern}(\frac{1}{2}+\gamma) \parallel \text{Bern}(\frac{1}{2}-\gamma)\right)} = \Omega\left(\frac{1}{\gamma^2}\right),$$

since the Bernoulli KL divergence is $\Theta(\gamma^2)$ as $\gamma \rightarrow 0$.

Theorem L.3 and Proposition L.4 cannot be combined into an implementable-policy optimality theorem: π^* receives the answer D_∞ as side information and lies outside the lower bound’s class. The exact Bellman characterization does not remove this information gap; it says what to optimize once a calibrated posterior model has been declared. Part 1 likewise makes no optimality claim for threshold aggregation.

Remark L.5 (Open implementable-policy comparison). To prove a matching rate for an implementable watcher or strategic-appellant policy, one must specify its prior or calibrated signal law, control spurious appeals after matching decisions, include finite depth, and use the deployed selection/reuse process. No implementable-policy rate-optimality result under the deployed process is claimed here.

L.4 Why the SPRT Analogy Is Limited

OD resets ballots rather than accumulating a likelihood ratio; stopping is chosen by agents; outcome categories are richer than binary hypotheses; and validator identities, incentives, and selection laws matter. Discrete geometric sizes are only one difference. Wald optimality therefore does not transfer to OD by a constant-factor argument.

L.5 Summary

Property	Status
Planner-optimal action for a declared finite-horizon economic model	✓Thm. A.5
Exact solution of one declared planner instance	✓Prop. A.8
Changing-column certificate composition and generator-shift transfer	✓Thm. A.10, Prop. A.12
One-round uniform and successive-PPSWOR panel bounds	✓Thms. C.4, C.6
Non-repeated nominal size-path overhead below $6\times$ an oracle adequate panel	✓Cor. A.14
Oracle benchmark cost $O(1/\gamma^2)$ when ladder reaches that scale	✓Thm. L.3
Vote-only sequential lower bound $\Omega(1/\gamma^2)$ at fixed error	✓Prop. L.4
Deployed OD optimality from combining those results	× Invalid: oracle side information and incentive gap
Finite-ladder watcher error bound	✓Thm. K.25
Implementable-policy optimality under deployed selection	Open

Supplementary Material

S1 Numerical Validation Details

The first pipeline evaluates a homogeneous independent-round benchmark over the statistical ladder. Votes are independent Bernoulli draws within and across model rounds, with reference-agreement probability $1/2 + \gamma$. It cross-checks exact round-zero errors, end-to-end oracle and watcher policies, expected evaluator work, the plug-in reversal score, and the top-round design margin. These are diagnostics of the stated model, not deployed-network estimates.

The second pipeline solves the four-round planner exactly and exercises the finite-population sampling stage omitted by the homogeneous benchmark. It fixes the heterogeneous population used in Section 3, computes the committee composition and mixed Poisson–Binomial tails exactly, and compares them with 200,000 Monte Carlo draws per displayed panel size. Figure 3 reports that primary check.

Neither pipeline simulates stake-weighted multi-round selection, validator reuse or replacement, a distinguished leader law, the minus-two appeal path, changing resolutions, capacity-dependent successors, wall-clock time, or endogenous multi-appellant behavior.

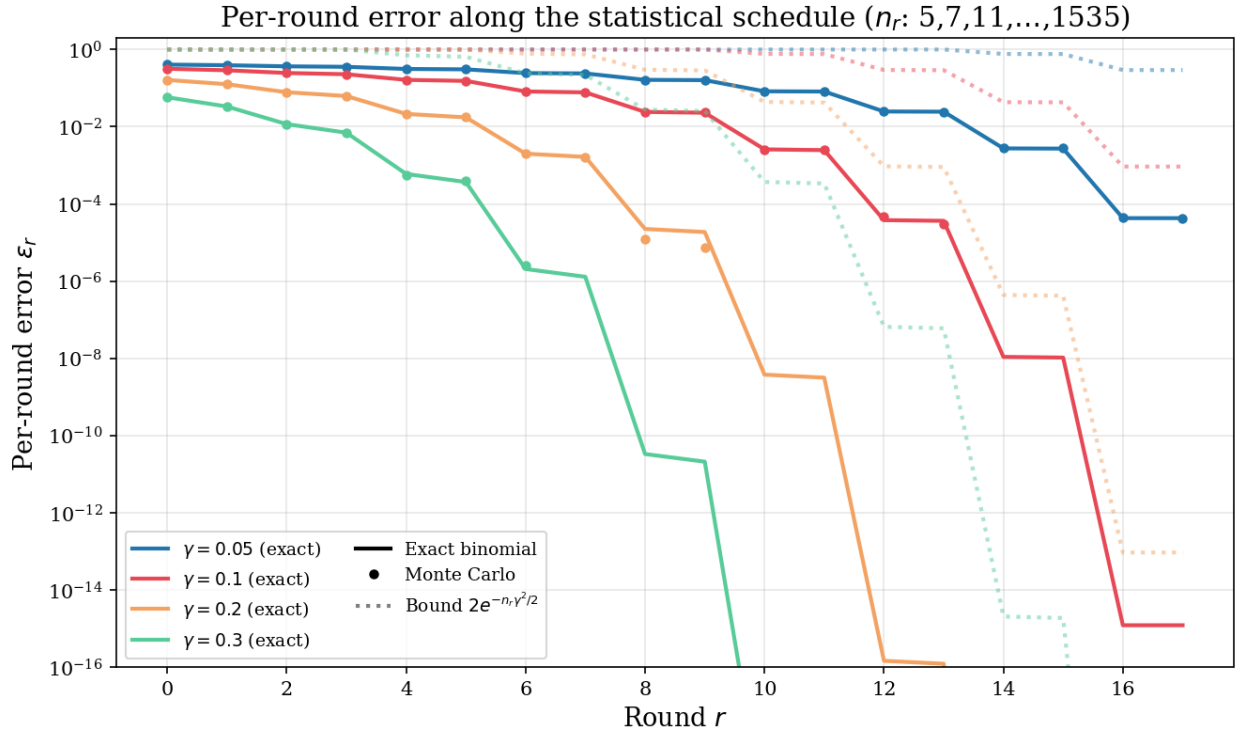


Figure S1. Per-round error along the statistical schedule for four homogeneous agreement margins. Solid lines are exact Binomial tails, circles are Monte Carlo, and dotted lines are the coarse two-stage bound. The bound dominates the exact error but is vacuous at small panels for moderate clarity.

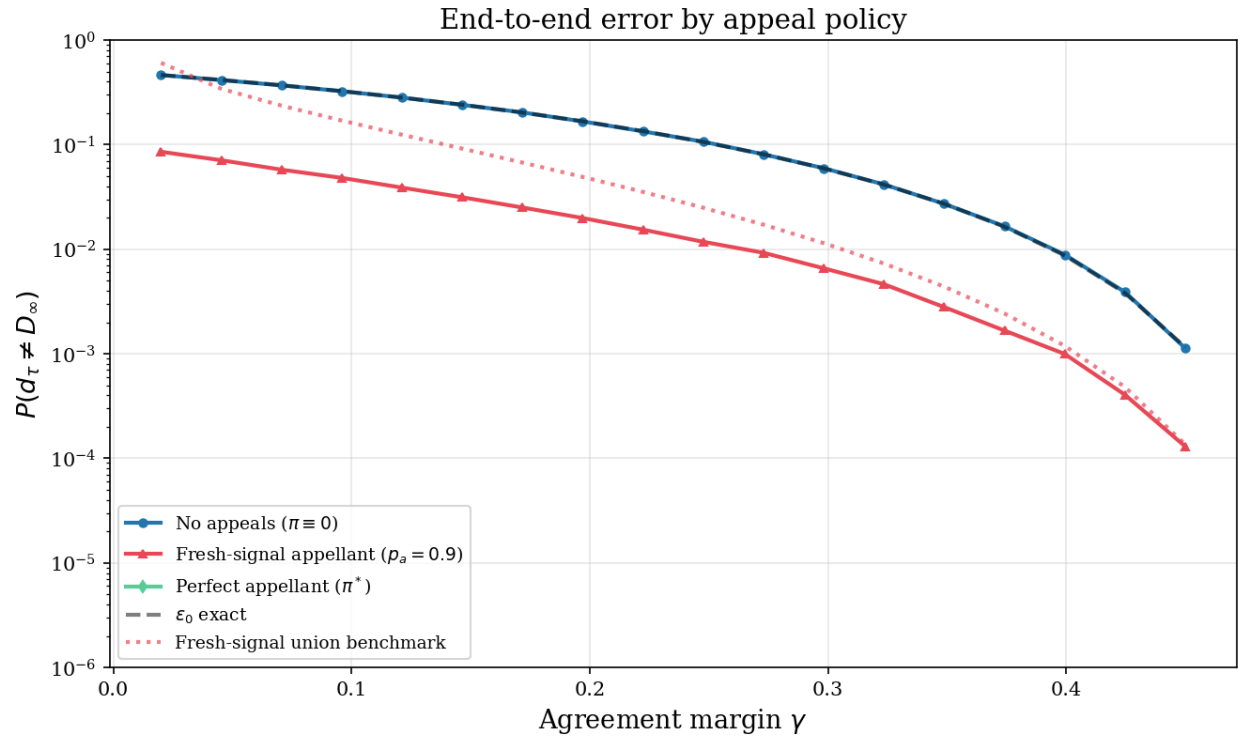


Figure S2. End-to-end disagreement probability by policy in the homogeneous independent-round benchmark. The fresh-signal watcher is calibrated by assumption; the oracle is an unimplementable lower benchmark.

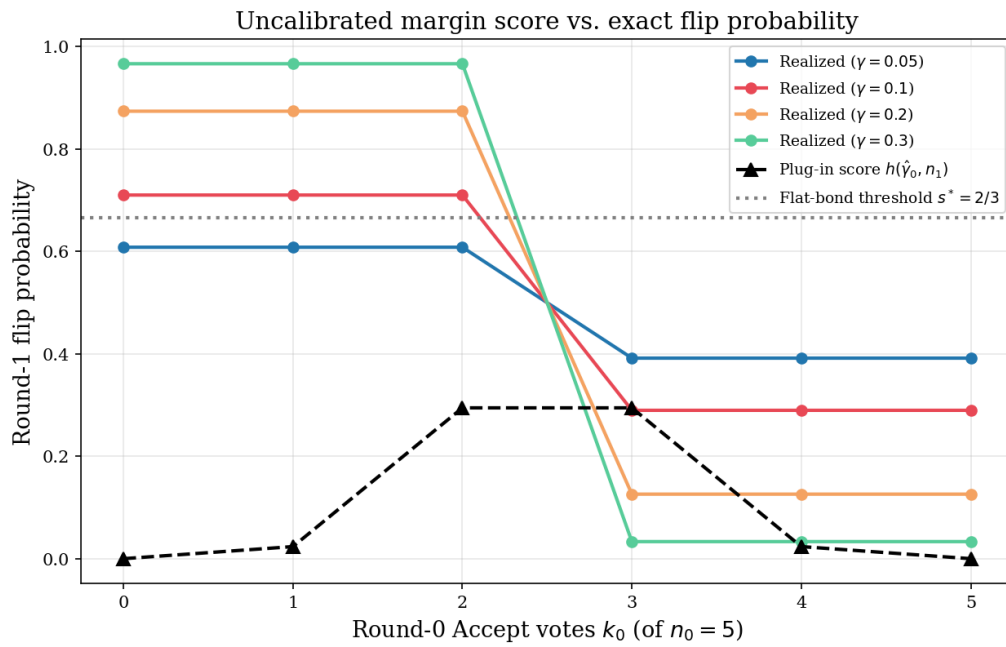


Figure S3. Round-one reversal probability by round-zero vote count. The black plug-in score is not an identified success probability: exact curves at different true margins disagree at the same observed count and are direction-sensitive.

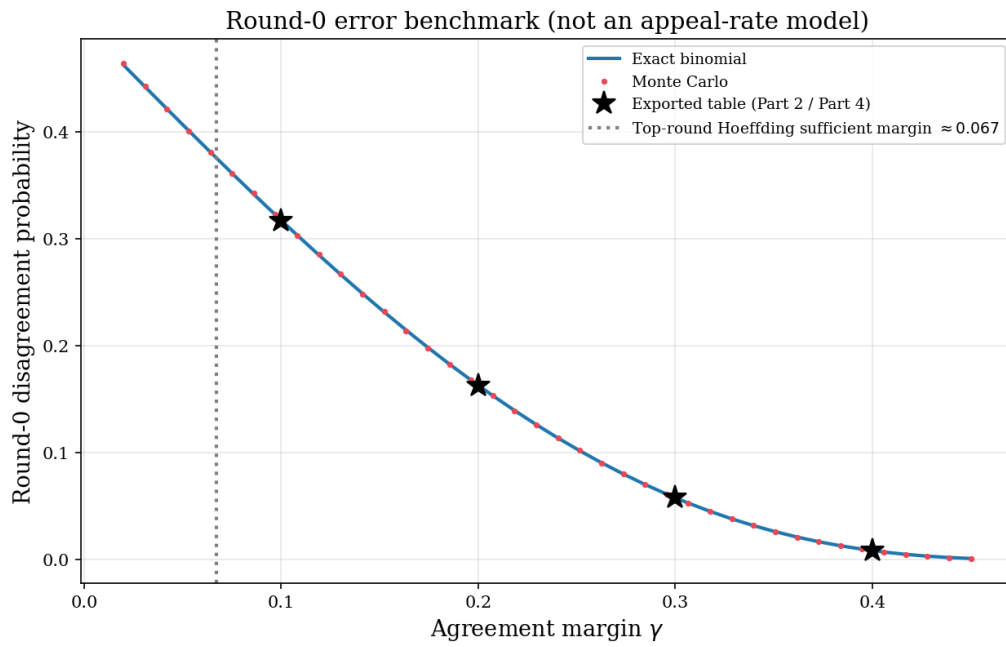


Figure S4. Round-zero disagreement probability under homogeneous agreement. This is an oracle appeal-pressure benchmark; a real appeal rate also depends on beliefs, private value, bonds, and protocol transitions.

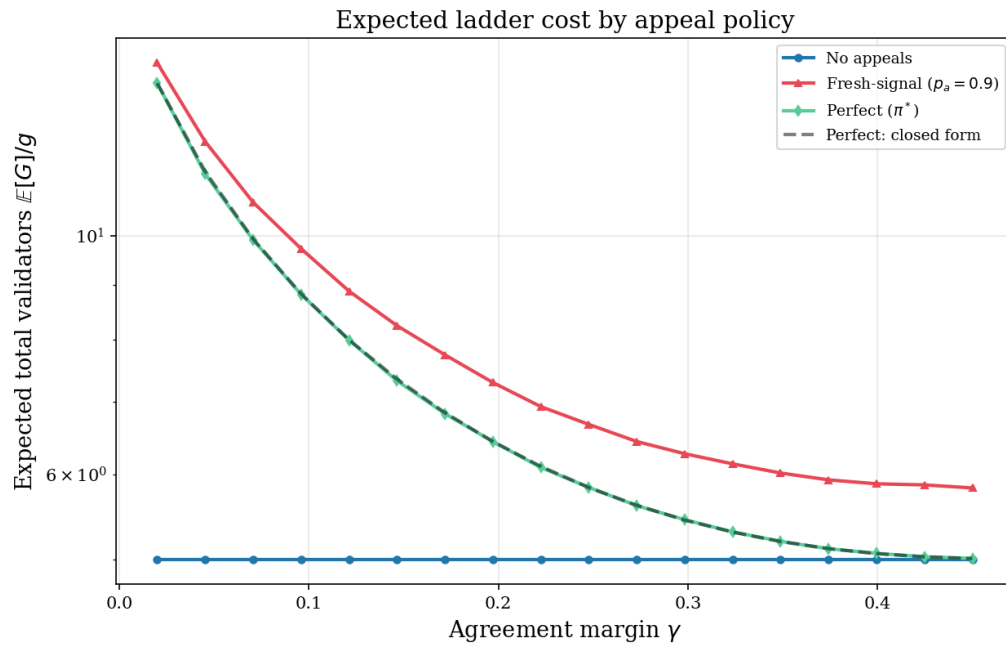


Figure S5. Expected model-ladder evaluator work by policy. The fresh-signal watcher can make both corrective and spurious appeals, while the oracle closed form matches Monte Carlo.

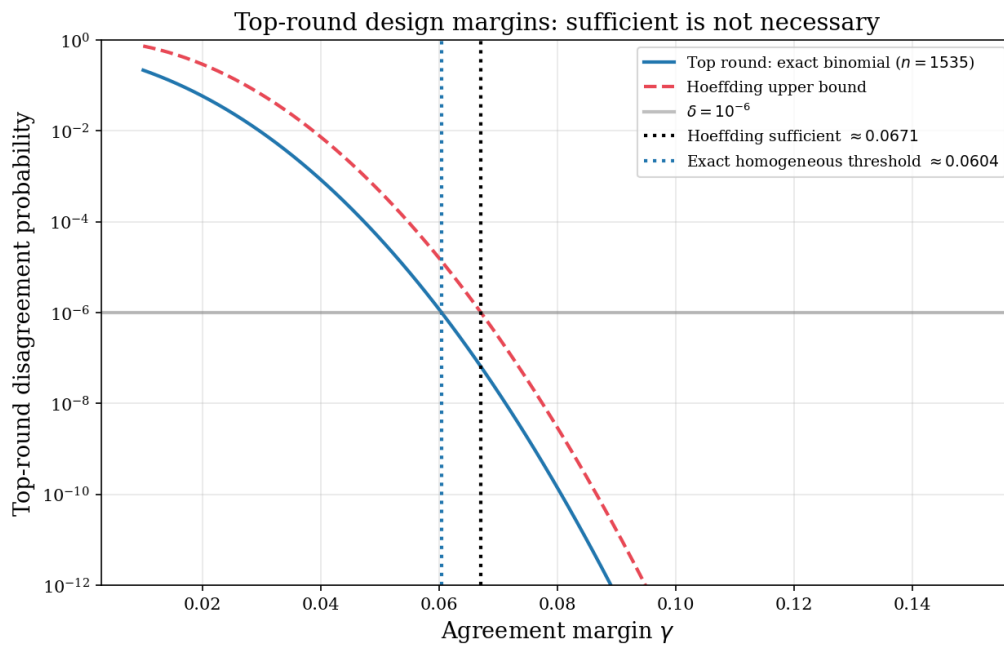


Figure S6. Top-round design margins at panel size 1535 and target error 10^{-6} . The exact homogeneous threshold is approximately 0.06043; the Hoeffding-sufficient value is 0.0671. Neither is an impossibility boundary for other laws or policies.

S2 Additional Numerical Interpretation

The homogeneous pipeline measures roughly 2.0% final disagreement for its fresh-signal watcher at agreement margin 0.2 and watcher agreement 0.9. This number depends on fresh independent rounds and a watcher whose signal quality is known. It must not be combined with the vote-only sequential lower bound to claim deployed OD optimality: the oracle and watcher policies possess side information unavailable to a vote-only stopping rule.

The weighted-without-replacement theorem is analytically valid but often underpowered. With a modeled population of 1543, full probability range, design margin 0.10, and several weight ratios, its optimized cap stays close to one through much of the ladder. Estimating inclusion probabilities from a stake snapshot is insufficient to repair this because the required design margin also depends on task-specific validator behavior. A useful deployment experiment must therefore predeclare a joint stake–performance model.

S3 Reproducibility

All displayed synthetic results are generated from the tracked scripts in `simulations/`. The homogeneous ladder pipeline uses NumPy random seed 42 and 200,000 or 400,000 replications depending on the diagnostic. The heterogeneous experiment uses seed 20260804 and 200,000 replications per panel size. Exact Binomial, Hypergeometric, and Poisson–Binomial calculations are evaluated with NumPy and SciPy; Matplotlib produces the figures.

From the paper directory, the complete validation is:

```
python3 simulations/ladder_validation.py
python3 simulations/structural_validation.py
python3 -m unittest discover -s simulations -p 'test_*.py'
```

The first two commands regenerate all figure files. The tests independently check the solved planner values and stopping pattern, finite-population means and inequalities, distinguished-leader sensitivity values, and the weighted-bound calibration table.

The source tree also records the package requirements and contains the rendered figures included in this PDF. The numerical campaign is entirely synthetic and does not require API credentials, validator identities, or private network data.